

A330

VER. 25.5.22

# KneeBoard



Click for Update

by Flyingdeuk

Domestic

Japan

China

SE Asia

Oceania

Mongo, ME Asia

Differences

FUEL Consumption

LP Aircon/ HP StartUnit

Cold Temp Correction

Cold Wx Operation

ENG ON Deicing

ENG OFF Deicing

QRH

RESET

# Domestic

**GMP**

**CJU**

**GMP**

**PUS**

**CJU**

**PUS**

**ICN**

**PUS**

Welcome PA

Next Page



## WELCOME PA (10분이상 지연 or 예상시 지연 방송)

손님 여러분, 안녕하십니까?

저는 여러분을 모시고 가는 기장 \_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_(국제)공항까지 비행시간은 \_\_시간 \_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,  
자리에 앉아 계실 때에는 항상 좌석벨트를  
매주시기 바랍니다. (저희 승무원 모두는)  
저는 여러분을 안전하게 모시기 위해 최선을  
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_(international)  
airport and our flight time is \_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

### Domestic

|     |      |
|-----|------|
| GMP | 김포국제 |
| ICN | 인천국제 |
| CJU | 제주국제 |
| PUS | 김해국제 |



도착방송

# 도착 방송 (5시간이상, 40분전)

출발지 기준 2200-0800 Quiet Hour

동남아 지역 난기류 대비 기내 서비스 절차

- 6시간 미만 중거리만 1시간 전 음료서비스
- 40분전 서비스 회수
- APP PA이후 CREW 착석 강조

손님 여러분, 저는 기장입니다.

우리 비행기는 앞으로 약 (40)분 후에

\_\_국제공항에 착륙 예정입니다.

현재 공항의 날씨는 ❶ \_\_, 기온은 섭씨 \_\_도 입니다.

❶ 맑으며

❶ (다소)흐리며

❶ (이슬)비가 내리며/소나기가 내리며

❶ 바람이 불고 있으며

❶ 눈이 오고 있으며

❶ 안개가 끼어 있으며

❶ 황사가 있으며

지금 이곳의 시각은 \_\_월 \_\_일 \_\_요일, 오전(오후)

\_\_시 \_\_분 입니다. 고맙습니다.

Ladies and gentlemen, this is the captain speaking.

We expect to land at \_\_ international airport in about (40) minutes.

The current temperature at \_\_ is \_\_ degrees Celsius, or \_\_ degrees Fahrenheit (NAVBLUE 참조)

and it is ❶ \_\_.

❶ (mostly) clear

❶ (partly) cloudy

❶ drizzling / raining

❶ windy

❶ snowing

❶ foggy

❶ hazy or smoggy

The current time is \_\_ : \_\_ a.m(p.m), on (day-of-the-week), (month)(date).

Thank you for flying with us today.



# Japan

GMP

HND

ICN

KIX

ICN

NRT

ICN

HND

ICN

FUK

ICN

NGO

Welcome PA

Next Page



# WELCOME PA (10분이상 지연 or 예상시 지연 방송)

손님 여러분, 안녕하십니까?

저는 여러분을 모시고 가는 기장 \_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_(국제)공항까지 비행시간은 \_\_시간 \_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,  
자리에 앉아 계실 때에는 항상 좌석벨트를  
매주시기 바랍니다. (저희 승무원 모두는)  
저는 여러분을 안전하게 모시기 위해 최선을  
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_(international)  
airport and our flight time is \_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

| Japan |                        |
|-------|------------------------|
| KIX   | 오사카/간사이                |
| HND   | 도쿄/하네다                 |
| NRT   | 도쿄/나리타                 |
| CTS   | 삿포로/신(New) 치토세         |
| NGO   | 나고야/주부(Chubu Centrair) |
| FUK   | 후쿠오카                   |



# China

ICN

HKG

ICN

PEK

ICN

WUH

ICN

CAN

ICN

XIY

Welcome PA

Next Page



# WELCOME PA (10분이상 지연 or 예상시 지연 방송)

손님 여러분, 안녕하십니까?

저는 여러분을 모시고 가는 기장 \_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_(국제)공항까지 비행시간은 \_\_시간 \_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,  
자리에 앉아 계실 때에는 항상 좌석벨트를  
매주시기 바랍니다. (저희 승무원 모두는)  
저는 여러분을 안전하게 모시기 위해 최선을  
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_(international)  
airport and our flight time is \_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

## China

HKG

홍콩

PEK

베이징/소우뚜(Capital)

WUH

우한/텐허

CAN

광저우/바이윈

XIY

시안/시엔양

TSN

텐진/빈하이

DLC

다롄/쑤우쑤이쯔





# SE Asia

PUS

BKK

ICN

BKK

ICN

SIN

ICN

DAD

ICN

SGN

ICN

CNX

Welcome PA

Next Page



# WELCOME PA (10분이상 지연 or 예상시 지연 방송)

손님 여러분, 안녕하십니까?

저는 여러분을 모시고 가는 기장 \_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_(국제)공항까지 비행시간은 \_\_시간 \_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,  
자리에 앉아 계실 때에는 항상 좌석벨트를  
매주시기 바랍니다. (저희 승무원 모두는)  
저는 여러분을 안전하게 모시기 위해 최선을  
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_(international)  
airport and our flight time is \_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

## S.E Asia

BKK

방콕/수완나폼

SIN

싱가포르/창이

DAD

다낭

SGN

베트남 호찌민/탄소넛

CNX

치앙마이

MNL

필리핀 마닐라/니노이 아키노

TPE

타이완/타이페이 타오유엔



**도착 방송**

# Oceania

NRT

HNL

ICN

SYD

ICN

BNE

Welcome PA

Next Page



# WELCOME PA (10분이상 지연 or 예상시 지연 방송)

손님 여러분, 안녕하십니까?

저는 여러분을 모시고 가는 기장 \_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_(국제)공항까지 비행시간은 \_\_시간 \_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,  
자리에 앉아 계실 때에는 항상 좌석벨트를  
매주시기 바랍니다. (저희 승무원 모두는)  
저는 여러분을 안전하게 모시기 위해 최선을  
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_(international)  
airport and our flight time is \_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

## Oceania

HNL

호놀룰루/다니엘 케이 이노우에

SYD

시드니/킹스폴드 스미스

BNE

브리즈번



도착방송

# Mongo, ME Asia

ICN

UBN

Welcome PA

Next Page



# WELCOME PA (10분이상 지연 or 예상시 지연 방송)

손님 여러분, 안녕하십니까?

저는 여러분을 모시고 가는 기장 \_\_입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

\_\_(국제)공항까지 비행시간은 \_\_시간 \_\_분  
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,  
자리에 앉아 계실 때에는 항상 좌석벨트를  
매주시기 바랍니다. (저희 승무원 모두는)  
저는 여러분을 안전하게 모시기 위해 최선을  
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for \_\_(international)  
airport and our flight time is \_\_ hours(s) and  
minutes.

For your safety, keep your seatbelts fastened  
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

## Mongo, ME Asia

UBN

울란바타르/칭기스칸



도착방송

# RKSS(GMP) 59ft | RKPC(CJU) 119ft

KE GMP 131.15      **PA**      KE CJU 129.4  
DCL -15분 가능 TOBT 5분 차이  
시 CTC Comm



Rwy 32R **Takeoff**  
(06:00L~0900L / 12:00L~15:00L  
/18:00L~21:00L)

GMP : **NADP FX-10도, CLB, CONF 1+F, Acc 4000ft**

|                     |                                    |                          |                                  |                          |
|---------------------|------------------------------------|--------------------------|----------------------------------|--------------------------|
| 32L/R               | <b>BULTI xT</b>                    | HD 324 ALT 5000 Mod NADP |                                  |                          |
|                     | (BULTI xQ)                         |                          |                                  |                          |
| 14L/R               | <b>BULTI xU</b>                    | HD 144 ALT 6000 Mod NADP |                                  |                          |
|                     | (BULTI xZ)                         |                          |                                  |                          |
| <b>KIP</b><br>113.6 | <b>32L IKMO</b><br>108.3           | <b>32R ISKP</b><br>110.7 | <b>14L ISEL</b><br>109.9         | <b>14R IOFR</b><br>108.7 |
| FIX                 | 32L/R : EO32L/R,<br>R225, YJU R271 |                          | 14L/R : EO14L/R,<br>R220, P73 /2 |                          |

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)



CJU **[066/246]** : STAR  
**AFT Merge PT(220kts) DCT IAF(210kts), FAF (160kts)**

|                                                  |                    |       |               |
|--------------------------------------------------|--------------------|-------|---------------|
| ILS Z 07                                         | DOTOL xP           | YUMIN | DOTOL 160     |
| <b>RNP Y 07 (No LPV), RNP Z 07 AR (RNP 0.11)</b> |                    |       |               |
| ILS Z 25                                         | DOTOL xT(xM)       | DUKAL | DOTOL/-10 160 |
| NAV                                              | <b>YDM 25 ICHE</b> |       |               |
| FIX                                              | RWxx /8            |       |               |

07 : P6(5176'), P5(5882'), P4(6840'-ATC HIRO)  
25 : P7(5219'), P8(5882'), P10(7524'-ATC HIRO)

Entering Rapid TWY CTC GND 121.675 (STOP x)  
HST 40KTS

|                        |                       |
|------------------------|-----------------------|
| <b>RKPC(CJU) 119ft</b> | <b>RKSS(GMP) 59ft</b> |
|------------------------|-----------------------|

KE CJU 129.4

DCL -10분

**PA**

KE GMP 131.15

Rwy 32L **Landing**(06:00L~0900L / 12:00L~15:00L  
/18:00L~21:00L)

CJU : SID (NADP 1)

07

KAMIT xE

HD 066 ALT 10000

25

KAMIT xW

HD 246 ALT 10000

YDM 109.0

07 109.9

25 ICHE 111.3

HUD

07 : NONE

25 : YDM246/3, R290

07 : Passing G4 CTC TWR

25 : 31 Holding PSN on P, E1,2,3 CTC TWR



GMP [324/144] : STAR

ILS 32L/R

OLMEN xT

BUMSI

OLMEN 160

ILS 14R

OLMEN xU

DOKDO

OLMEN 160

NAV

KIP 32L IKMO, 32R ISKP, 14L ISEL, 14R IOFR

FIX

KIP /8(RWY 32), YJU R271, P73 /2

32L : D3(6532'), E2(9117'), 32R(No CL L/T) : E1(6614')  
14R : C1(6578')

32L/R : 8 KIP L/G, 14R : LOC CAPT L/G

FAF : Final Flap

TWR -&gt; GND -&gt; APRON (All by ATC)

Except RWY14R Landing (Until R)



**RKSS(GMP) 59ft** **RKPK(PUS) 13ft**

KE GMP 131.15 **PA** KE Gimhae 129.2  
DCL -15분 가능 TOBT 5분 차이  
시 CTC Comm



Rwy 32R **Takeoff**  
(06:00L~0900L / 12:00L~15:00L  
/18:00L~21:00L)

GMP : **NADP FX-10도, CLB, CONF 1+F, Acc 4000ft**

|              |                                    |                           |                                  |                   |
|--------------|------------------------------------|---------------------------|----------------------------------|-------------------|
| 32L/R        | OSPOT xT                           | HD 324 ALT 5000 Modi NADP |                                  |                   |
|              | (OSPOT xQ)                         |                           |                                  |                   |
| 14L/R        | OSPOT xU                           | HD 144 ALT 6000 Modi NADP |                                  |                   |
|              | (OSPOT xZ)                         |                           |                                  |                   |
| KIP<br>113.6 | 32L IKMO<br>108.3                  | 32R ISKP<br>110.7         | 14L ISEL<br>109.9                | 14R IOFR<br>108.7 |
| HUD          | 32L/R : EO32L/R,<br>R225, YJU R271 |                           | 14L/R : EO14L/R,<br>R220, P73 /2 |                   |

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)



PUS [002/182] : STAR (T/W 10tks, 36R, 360000lbs 제한)

|        |                         |       |                                   |
|--------|-------------------------|-------|-----------------------------------|
| ILS 36 | KEVOX x                 | MASTA | 9DME LG, 8DME FLAP                |
| VOR 18 | GAYHA x                 | MASTA | <b><u>18 Circling Click!!</u></b> |
| NAV    | KMH, 36L IKMA, 36R IKHE |       |                                   |
| FIX    | 36 : IKMA/IKHE /9, /8   |       | 18 : KMH R284, R280               |

36L : **C4 (6299')**, C2(7795') 36R : E3(5866'), E2(7339')  
18R(DIS):C6(5770'), C7(6824') 18L:E4(5882'), E5(8792')

Vacate **C3, C4** by ATC only. Max Taxi SPD 20KTS  
C2 HOLD SHORT 가까움(Vacate TaxiSPD)

**RKPK(PUS) 13ft****RKSS(GMP) 59ft**

KE Gimhae 129.2

DCL -5분

**PA**

KE GMP 131.15

Rwy 32L **Landing**(06:00L~0900L / 12:00L~15:00L  
/18:00L~21:00L)**PUS : NADP CLB 1500, 360000lbs TOGA, CONF 1+F**

36

**SOORO x  
KALOD tx**HD 002 **ALT ATC** **Modi NADP**

18

**GIMHAE x**HD 182 **ALT 5000** **Modi NADP****KMH 113.8****PSN 114.0****36L IKMA  
108.5****36R IKHE  
109.5**

FIX

36 : KMH R091, R271, R185

RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS

**GMP [324/144] : STAR**

ILS 32L/R

GUKDO xT

BUMSI

GUKDO 160

ILS 14R

GUKDO xU

DOKDO

GUKDO 160

NAV

**KIP 32L IKMO, 32R ISKP, 14L ISEL, 14R IOFR**

FIX

KIP /8(RWY 32), YJU R271, P73 /2

**32L : D3(6532'), E2(9117'), 32R (No CL L/T) : E1(6614')**  
**14R : C1(6578')**

32L/R : 8 KIP L/G, 14R : LOC CAPT L/G

FAF : Final Flap

TWR -&gt; GND -&gt; APRON (All by ATC)

Except RWY14R Landing (Until R)

|                        |                              |
|------------------------|------------------------------|
| <b>RKPC(CJU) 119ft</b> | <b><u>RKPK(PUS) 13ft</u></b> |
|------------------------|------------------------------|

KE CJU 129.4

DCL -10분

**PA**

KE Gimhae 129.2

CJU : SID (NADP 1)

07

AKPON xE

HD 066 ALT 9000

25

AKPON xW

HD 246 ALT ATC

YDM 109.0

07 109.9

25 ICHE 111.3

FIX

07 : NONE

25 : YDM246/3, R290

07 : Passing G4 CTC TWR

25 : 31 Holding PSN on P, E1,2,3 CTC TWR



PUS [002/182] : STAR (T/W 10tks, 36R, 360000lbs 제한)

ILS 36

KEVOX x

ANROD

9DME LG, 8DME FLAP

VOR 18

GAYHA x

ANROD

**18 Circling Click!!**

NAV

KMH, 36L IKMA, 36R IKHE

FIX

36 : IKMA/IKHE /9, /8

18 : KMH R284, R280

36L : **C4 (6299')**, C2(7795') / 36R : E3(5866'), E2(7339')

18R(DIS):C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate **C3,C4** by ATC only. Max Taxi SPD 20KTS

C2 HOLD SHORT 가까움(Vacate TaxiSPD)

**RKPK(PUS) 13ft**

**RKPC(CJU) 119ft**

KE Gimhae 129.2

DCL -5분

**PA**

KE CJU 129.4

PUS : NADP CLB 1500, 360000lbs TOGA, CONF 1+F

36

SOORO x  
TOPAX tx

HD 002 ALT ATC Modi NADP

18

BULIM x  
ENGOT tx

HD 182 ALT 5000 Modi NADP

KMH 113.8

PSN 114.0

36L IKMA  
108.5

36R IKHE  
109.5

FIX

36 : KMH R091, R271, R185

RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS



CJU [066/246] : STAR

AFT Merge PT(220kts) DCT IAF(210kts), FAF (160kts)

ILS Z 07

UPGOS xP

YUMIN

RNP Y 07 (No LPV), RNP Z 07 AR (RNP 0.11)

ILS Z 25

UPGOS xT(xM)

DUKAL

NAV

YDM 25 ICHE

FIX

RWxx /8

07 : P6(5176'), P5(5882'), P4(6840'-ATC HIRO)

25 : P7(5219'), P8(5882'), P10(7524'-ATC HIRO)

Entering Rapid TWY CTC GND 121.675, STOP X  
HST 40KTS

|                                                 |                                   |                               |                                  |                    |
|-------------------------------------------------|-----------------------------------|-------------------------------|----------------------------------|--------------------|
| <u>RKSI(ICN) 23ft</u>                           |                                   | <u>RKPK(PUS) 13ft</u>         |                                  |                    |
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm |                                   | PA KE Gimhae 129.2            |                                  |                    |
| ICN : SID (33/34 NADP 1, 15/16 NADP 2)          |                                   |                               |                                  |                    |
| 33L/R                                           | OSPOT xE/A                        | HD 333 ALT 5500/ATC SPD 230/x |                                  |                    |
| 34L/R                                           | OSPOT xY                          | HD 333 ALT ATC SPD 230        |                                  |                    |
| 15L/R                                           | OSPOT xC                          | HD 153 ALT 5000 NADP2         |                                  |                    |
| 16L/R                                           | OSPOT xH                          | HD 153 ALT 5000 NADP2         |                                  |                    |
| NCN<br>113.8                                    | 33L INLL<br>109.3                 | 33R INRR<br>108.9             | 15L ISLL<br>111.9                | 15R ISRR<br>109.1  |
| WNG<br>112.9                                    | 34L IRFN<br>109.95                | 34R IRKN<br>108.1             | 16L IRKS<br>110.35               | 16R IRFS<br>108.55 |
| FIX                                             | 33L/R : NC05L/R, R242<br>YJU R271 |                               | 34L/R : EO34L/R,R242<br>YJU R271 |                    |
| Parallel TWY 10KTS 이상(R17 MAX 15kts)            |                                   |                               |                                  |                    |



|                                                     |                             |       |                            |
|-----------------------------------------------------|-----------------------------|-------|----------------------------|
| PUS [002/182] : STAR (T/W 10tks, 36R, 360000lbs 제한) |                             |       |                            |
| ILS 36                                              | KEVOX x                     | MASTA | 9DME LG, 8DME FLAP         |
| VOR 18                                              | GAYHA x                     | MASTA | <u>18 Circling Click!!</u> |
| NAV                                                 | KMH KMH, 36L IKMA, 36R IKHE |       |                            |
| FIX                                                 | 36 : IKMA/IKHE /9, /8       |       | 18 : KMH R284, R280        |

36L : C4 (6299'), C2(7795') / 36R : E3(5866'), E2(7339')  
18R(DIS): C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate C3,C4 by ATC only, Max Taxi SPD 20KTS  
C2 HOLD SHORT 가까움(Vacate TaxiSPD)

**RKPK(PUS) 13ft****RKSI(ICN) 23ft**

KE Gimhae 129.2

DCL -5분

**PA**

KE ICN 131.5

**PUS : NADP CLB 1500, 360000lbs TOFA, CONF 1+F**

36

SOORO x  
KALOD txHD 002 ALT ATC  
Modi NADP

18

GIMHAE x

HD 182 ALT 5000 Modi NADP

KMH 113.8

PSN 114.0

36L IKMA  
108.536R IKHE  
109.5

FIX

36 : KMH R091, R271, R185

RWY 36 400ft Man L/H turn, Max Taxi SPD 20KTS



ICN [333/153] : STAR

33/34

GUKDO xE

ENPIL

GUKDO 180

15/16

GUKDO xH

MUNAN

GUKDO 180

NAV

NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR

WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS

FIX

RWY /8(180kts), /5(160kts), YJU R271

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')

15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')

16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO

RKSS(GMP) 59ft

RJTT(HND) 21ft

KE GMP 131.15  
DCL -15분 가능  
TOBT5분 차이시 CTC Comm

PA

Delta Oper 132.075



Rwy 32R **Takeoff**  
(06:00L~0900L / 12:00L~15:00L  
/18:00L~21:00L)

GMP : **NADP FX-10도, CLB, CONF 1+F, Acc 4000ft**

|              |                                    |                           |                                  |                   |
|--------------|------------------------------------|---------------------------|----------------------------------|-------------------|
| 32L/R        | EGOBA xT                           | HD 324 ALT 5000 Modi NADP |                                  |                   |
|              | (EGOBA xQ)                         |                           |                                  |                   |
| 14L/R        | EGOBA xU                           | HD 144 ALT 6000 Modi NADP |                                  |                   |
| KIP<br>113.6 | 32L IKMO<br>108.3                  | 32R ISKP<br>110.7         | 14L ISEL<br>109.9                | 14R IOFR<br>108.7 |
| HUD          | 32L/R : EO32L/R,<br>R225, YJU R271 |                           | 14L/R : EO14L/R,<br>R220, P73 /2 |                   |

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)



DEP 125.15 – TGU 134.17 – FUK 133.02

TKO 120.5 – TKO 133.35

TKO APP 119.1 – 119.65



HND : **STAR XAC Night– APP xxx Y 1400z~ SPENS 220**  
**22/23 LDA, VOR 16L/R Tailored Procedure**

|         |                                                      |             |                                            |
|---------|------------------------------------------------------|-------------|--------------------------------------------|
| 34L/R   | XAC xK/H                                             | KAIHO/CACAO | ILS X / VIS                                |
| 22[222] | XAC 1B                                               | BACON       | LDA W(RNV22-W)                             |
| 16R/L   | XAC R                                                | NATTY/SANDY | RNP(RNV16R/L)                              |
| 23[232] | XAC 2B                                               | DANON       | LDA W(RNV23-W)                             |
| NAV     | HME, 34L IHA, 34R ITC(LDA chk:22 IKL,23 ITL)         |             |                                            |
| FIX     | 22 : IKL /8, /3, HME 015<br>23 : ITL /12,/7, HME 177 |             | Z 34L : IHA /10, /5<br>Z 34R : ITC /10, /5 |

34L:L12(6515),L13(7165), 16R(DIS):L5(5147),L3(6361)  
22 : B4(6207), B3(6830), 23: D5(5072),D3(6391)  
34R(DIS):C10(5780),C11(7270), 16L(DIS):C6(5925),C4(7004)

xxx Z : 180kts, 160kts limit APP Chart, xxx Y After 1400z

Gate 105P Procedure

[LDA22/23 Next Page]

# A330 HND LDA 22 W

F-PLN

APPR

STAR

TRANS

RNV22-W

XAC 1B

BACON

CHK IKL, IKL/8 180kts, /3 160kts, HME 015 (GA)



고려사항

- Procedure SPD 지켜야 Traffic 간격 유지됨. 무게에 따라 -4도 증속 가능성 있음.
- S/B 사용 하거나 강하 각을 조절 하는 것이 필요함.

## Missed App

R/H Turn HME 015 OUTBD (PULL TRK)

(DCT KASGA Radial In 196, MAN NAV)

# A330 HND LDA 23 W

F-PLN

APPR

STAR

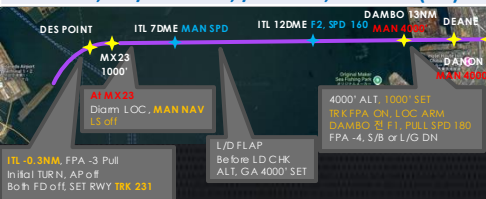
TRANS

RNV23-W

XAC 2B

BACON

CHK ITL, ITL/12 180kts, /7 160kts, HME 177 (GA)



## Missed App

L/H Turn HME 177 OUTBD (PULL TRK)

(DCT UTIBO Radial In 357, MAN NAV)



# RJTT(HND) 21ft | RKSS(GMP) 59ft

Delta Oper 132.075 **PA** KE GMP 131.15  
DCL -15분

Rwy 32L **Landing**  
(06:00L~0900L / 12:00L~15:00L  
/18:00L~21:00L)



HND : SID (xx B/C 2200-0230z 0600-1000z) NADP 1

|     |                    |                                                      |
|-----|--------------------|------------------------------------------------------|
| ALL | BEKLA x<br>OPPAR x | HD 338/158,231/051,223/043<br>ALT ATC<br>Mod SPD 230 |
|-----|--------------------|------------------------------------------------------|

|              |                  |                   |                  |                   |                 |                 |
|--------------|------------------|-------------------|------------------|-------------------|-----------------|-----------------|
| HME<br>112.2 | 34L IHA<br>111.7 | 16R ITA<br>111.55 | 34R ITC<br>108.9 | 16L IOC<br>111.95 | 22 IAD<br>108.1 | 23 ITD<br>110.5 |
|--------------|------------------|-------------------|------------------|-------------------|-----------------|-----------------|

|     |                                                                      |
|-----|----------------------------------------------------------------------|
| FIX | 34L : HME 351/1.1, R095, 34R : HME R080,<br>R095, 22 : HME /2.2 R185 |
|-----|----------------------------------------------------------------------|

34R BEKLA : KAIJI 230kts, TO FLAP SPD Until TORAM  
05 OPPAR : Reduce SPD ARC TURN  
RWY05 RTE5 TAXI Chart



DEP ATIS

TKO 120.5 – FUK 133.02

TGU 120.57

APP 119.75



GMP [324/144] : STAR

|           |          |       |           |
|-----------|----------|-------|-----------|
| ILS 32L/R | GUkDO xT | BUMSI | GUkDO 160 |
| ILS 14R   | GUkDO xU | DOKDO | GUkDO 160 |

|     |                                            |
|-----|--------------------------------------------|
| NAV | KIP 32L IKMO, 32R ISKP, 14L ISEL, 14R IOFR |
|-----|--------------------------------------------|

|     |                                  |
|-----|----------------------------------|
| FIX | KIP /8(RWY 32), YJU R271, P73 /2 |
|-----|----------------------------------|

32L : D3(6532'), E2(9117'), 32R (No CL L/T) : E1(6614')  
14R : C1(6578')

32L/R : 8 KIP L/G, 14R : LOC CAPT L/G  
FAF : Final Flap  
TWR -> GND -> APRON (All by ATC)  
Except RWY14R Landing (Until R)

|                              |                              |
|------------------------------|------------------------------|
| <b><u>RKSI(ICN) 23ft</u></b> | <b><u>RJBB(KIX) 17ft</u></b> |
|------------------------------|------------------------------|

|                                                  |           |               |
|--------------------------------------------------|-----------|---------------|
| KE ICN 131.5<br>DCL -10분 TOBT 5분 차이시<br>CTC Comm | <b>PA</b> | KE KIX 130.95 |
|--------------------------------------------------|-----------|---------------|

**ICN : SID (33/34 NADP 1, 15/16 NADP 2)**

|              |                                   |                               |                                   |                    |
|--------------|-----------------------------------|-------------------------------|-----------------------------------|--------------------|
| 33L/R        | EGOBA xE/A                        | HD 333 ALT 5500/ATC SPD 230/x |                                   |                    |
| 34L/R        | EGOBA xY                          | HD 333 ALT ATC SPD 230        |                                   |                    |
| 15L/R        | EGOBA xC                          | HD 153 ALT 5000 NADP2         |                                   |                    |
| 16L/R        | EGOBA xH                          | HD 153 ALT 5000 NADP2         |                                   |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9             | 15L ISLL<br>111.9                 | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1             | 16L IRKS<br>110.35                | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                               | 34L/R : EO34L/R, R242<br>YJU R271 |                    |

Parallel TWY 10KTS 이상(R17 MAX 15kts)

|                                                      |
|------------------------------------------------------|
| <a href="#">DEP 125.15 – TGU 134.17 – FUK 124.15</a> |
| <a href="#">TKO 133.8</a>                            |
| <a href="#">KIX RDR 120.85</a>                       |
| <a href="#">KIX APP 120.25</a>                       |



**KIX [058/238] : STAR (SAEKI 170, RANDY 150)**

|              |                                                |              |                    |
|--------------|------------------------------------------------|--------------|--------------------|
| <b>06L</b>   | <b>ALISA B</b>                                 | <b>BERRY</b> | <b>ILS Y 06L</b>   |
| <b>06R</b>   | <b>ALISA A</b>                                 | <b>ALLAN</b> | <b>ILS Y 06R</b>   |
| <b>24L/R</b> | <b>ALISA C</b>                                 | <b>MAYAH</b> | <b>ILS Z 24L/R</b> |
| <b>NAV</b>   | <b>KIE, 06L IKJ, 06R IKD, 24L IKN, 24R IKW</b> |              |                    |
| <b>FIX</b>   | <b>RWxx /8</b>                                 |              |                    |

06L : B8(5160'), B6(6751'), 24R : B7(5318'), B9(6751')  
06R : A7(5137'), A6(6938'), 24L : A8(5269'), A9(6976')

RWY06 : After 2500ft L/G DN, After 1500ft L/D FLAP  
TAXI RTE 1(via J4), 2(via J3)

RJBB(KIX) 17ft

RKSI(ICN) 23ft

KE KIX 130.95

DCL -15분

**PA**

KE ICN 131.5

KIX : SID – SOUJA tx (NADP 1)

06L/R

HELEN x

HD 059 ATC (9000)

24L/R

- SOUJA tx

HD 239 ATC (9000) SPD 210

KIE

111.6

06L IKJ

108.7

06R IKD

108.1

24L IKN

110.7

24R IKW

108.5

APU Start, TAXI RTE 1(via J4), 2(via J3)

[DEP 119.2](#)

[TKO 132.7 – 133.8](#)

[FUK 124.15](#)

[TGU 120.57](#)

[APP 119.75](#)



ICN [333/153] : STAR

33/34

GUkDO xE

ENPIL

GUkDO 180

15/16

GUkDO xH

MUNAN

GUkDO 180

NAV

NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR

WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS

FIX

RWY /8(180kts), /5(160kts), YJU R271

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')

15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')

16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO

|                       |                        |
|-----------------------|------------------------|
| <b>RKSI(ICN) 23ft</b> | <b>RJAA(NRT) 135ft</b> |
|-----------------------|------------------------|

|                                                 |                           |
|-------------------------------------------------|---------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> KE Tokyo 131.70 |
|-------------------------------------------------|---------------------------|

|                                               |
|-----------------------------------------------|
| <b>ICN : SID (33/34 NADP 1, 15/16 NADP 2)</b> |
|-----------------------------------------------|

|                     |                                   |                                      |                                  |                           |
|---------------------|-----------------------------------|--------------------------------------|----------------------------------|---------------------------|
| <b>33L/R</b>        | <b>EGOBA xE/A</b>                 | <b>HD 333 ALT 5500/ATC SPD 230/x</b> |                                  |                           |
| <b>34L/R</b>        | EGOBA xY                          | HD 333 ALT ATC SPD 230               |                                  |                           |
| <b>15L/R</b>        | <b>EGOBA xC</b>                   | <b>HD 153 ALT 5000 NADP2</b>         |                                  |                           |
| <b>16L/R</b>        | EGOBA xH                          | HD 153 ALT 5000 NADP2                |                                  |                           |
| <b>NCN</b><br>113.8 | <b>33L INLL</b><br>109.3          | <b>33R INRR</b><br>108.9             | <b>15L ISLL</b><br>111.9         | <b>15R ISRR</b><br>109.1  |
| <b>WNG</b><br>112.9 | <b>34L IRFN</b><br>109.95         | <b>34R IRKN</b><br>108.1             | <b>16L IRKS</b><br>110.35        | <b>16R IRFS</b><br>108.55 |
| <b>FIX</b>          | 33L/R : NC05L/R, R242<br>YJU R271 |                                      | 34L/R : EO34L/R,R242<br>YJU R271 |                           |

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

|                                                                                                                                        |                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <a href="#">DEP 125.15 – TGU 134.17</a><br><a href="#">TKO 124.15 – 132.02 – 124.1– 128.2</a><br><a href="#">TKO APP 124.4 – 120.2</a> |  |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

|                                                      |
|------------------------------------------------------|
| <b>NRT : HAKKA 330,YAGAN 240,LIVET 210,SWAMP 150</b> |
|------------------------------------------------------|

|                   |                                                                            |                         |                     |
|-------------------|----------------------------------------------------------------------------|-------------------------|---------------------|
| <b>34L/R[337]</b> | <b>SWAMP E</b><br>(SWAMP T)                                                | <b>ELGAR</b><br>(TYLER) | <b>ILS 34L/R(Z)</b> |
| <b>16L/R[157]</b> | <b>SWAMP G</b><br>(SWAMP N)                                                | <b>GEMIN</b><br>(NORMA) | <b>ILS Z 16L/R</b>  |
| <b>NAV</b>        | <b>NRE, 16L ITM, 16R IKF, 34L IYQ, 34R ITJ</b>                             |                         |                     |
| <b>FIX</b>        | 16L : ITM 4 / 34R : ITJ 14, 4 (DME)<br>16R : IKF 4 / 34L : IYQ 12, 4 (DME) |                         |                     |

|                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------|
| <b>16L : B6(6433'), B7(7017'), 34R : B4(5849'), B2(6778')</b><br><b>16R : A6(6076'), A7(7624'), 34L : A5(6167'), A4(7641')</b> |
|--------------------------------------------------------------------------------------------------------------------------------|

|                                                                                          |
|------------------------------------------------------------------------------------------|
| L/D DOWN before 14/12 DME, L/D FLAP 4 DME<br>Arrival Taxi RTE in Jeppesen (No Numbering) |
|------------------------------------------------------------------------------------------|

|                        |                       |
|------------------------|-----------------------|
| <u>RJAA(NRT) 135ft</u> | <u>RKSI(ICN) 23ft</u> |
|------------------------|-----------------------|

KE Tokyo 131.70  
DCL -15분

**PA**

KE ICN 131.5

NRT : SID – ENPAR tx (NADP 1)

|              |                                |                     |                  |                  |  |
|--------------|--------------------------------|---------------------|------------------|------------------|--|
| 16L/R        | TETRA x<br>ENPAR tx<br>CHK ALT | HD 158 ALT ATC      |                  |                  |  |
| 34L/R        |                                | HD 338 ALT 7000/ATC |                  |                  |  |
| NRE<br>117.9 | 16L ITM<br>110.7               | 16R IKF<br>111.5    | 34L IYQ<br>111.9 | 34R ITJ<br>110.9 |  |

APU Start, TAXI RTE 1, 2, 3, 4 RWY 별 DEP RTE

[DEP 124.2](#)

[TKO 120.5 – 133.45 – 133.02 – 133.8](#)

[TGU 120.57](#)

[APP 119.75](#)



ICN [333/153] : STAR

|       |                                             |       |           |
|-------|---------------------------------------------|-------|-----------|
| 33/34 | GUkDO xE                                    | ENPIL | GUkDO 180 |
| 15/16 | GUkDO xH                                    | MUNAN | GUkDO 180 |
| NAV   | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR |       |           |
|       | WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |       |           |
| FIX   | RWY /8(180kts), /5(160kts), YJU R271        |       |           |

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')  
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')  
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO

| RKSI(ICN) 23ft                                                                                                                                                    |                                                       |                   | RJTT(HND) 21ft                             |                                                                                      |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------|--------------------------------------------|--------------------------------------------------------------------------------------|--|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm                                                                                                                   |                                                       |                   | PA<br>Delta Oper 132.075                   |                                                                                      |  |
| ICN : SID (33/34 NADP 1, 15/16 NADP 2)                                                                                                                            |                                                       |                   |                                            |                                                                                      |  |
| 33L/R                                                                                                                                                             | EGOBA xE/A                                            |                   | HD 333 ALT 5500/ATC SPD 230/x              |                                                                                      |  |
| 34L/R                                                                                                                                                             | EGOBA xY                                              |                   | HD 333 ALT ATC SPD 230                     |                                                                                      |  |
| 15L/R                                                                                                                                                             | EGOBA xC                                              |                   | HD 153 ALT 5000 NADP2                      |                                                                                      |  |
| 16L/R                                                                                                                                                             | EGOBA xH                                              |                   | HD 153 ALT 5000 NADP2                      |                                                                                      |  |
| NCN<br>113.8                                                                                                                                                      | 33L INLL<br>109.3                                     | 33R INRR<br>108.9 | 15L ISLL<br>111.9                          | 15R ISRR<br>109.1                                                                    |  |
| WNG<br>112.9                                                                                                                                                      | 34L IRFN<br>109.95                                    | 34R IRKN<br>108.1 | 16L IRKS<br>110.35                         | 16R IRFS<br>108.55                                                                   |  |
| FIX                                                                                                                                                               | 33L/R : NC05L/R, R242<br>YJU R271                     |                   | 34L/R : EO34L/R, R242<br>YJU R271          |                                                                                      |  |
| Parallel TWY 10KTS 이상(R17 MAX 15kts)                                                                                                                              |                                                       |                   |                                            |                                                                                      |  |
| DEP 125.15 – TGU 134.17 – FUK 133.02<br>TKO 120.5 – TKO 133.35<br>TKO APP 119.1 – 119.65                                                                          |                                                       |                   |                                            |  |  |
| HND : STAR XAC Night– APP xxx Y 1400z~ SPENS 220<br>22/23 LDA, VOR 16L/R Tailored Procedure                                                                       |                                                       |                   |                                            |                                                                                      |  |
| 34L/R                                                                                                                                                             | XAC xK/H                                              | KAIHO/CACAO       | ILS X / VIS                                |                                                                                      |  |
| 22[223]                                                                                                                                                           | XAC 1B                                                | BACON             | LDA W(RNV22-W)                             |                                                                                      |  |
| 16R/L                                                                                                                                                             | XAC R                                                 | NATTY/SANDY       | RNP(RNV16R/L)                              |                                                                                      |  |
| 23[231]                                                                                                                                                           | XAC 2B                                                | DANON             | LDA W(RNV23-W)                             |                                                                                      |  |
| NAV                                                                                                                                                               | HME, 34L IHA, 34R ITC(LDA chk:22 IKL, 23 ITL)         |                   |                                            |                                                                                      |  |
| FIX                                                                                                                                                               | 22 : IKL /8, /3, HME 015<br>23 : ITL /12, /7, HME 177 |                   | Z 34L : IHA /10, /5<br>Z 34R : ITC /10, /5 |                                                                                      |  |
| 34L:L12(6515),L13(7165), 16R(DIS):L5(5147),L3(6361)<br>22 : B4(6207), B3(6830), 23: D5(5072),D3(6391)<br>34R(DIS):C10(5780),C11(7270), 16L(DIS):C6(5925),C4(7004) |                                                       |                   |                                            |                                                                                      |  |
| xxx Z : 180kts, 160kts limit APP Chart, xxx Y After 1400z                                                                                                         |                                                       |                   |                                            |                                                                                      |  |
| Gate 105P Procedure                                                                                                                                               |                                                       |                   | [LDA22/23 Next Page]                       |                                                                                      |  |

# A330 HND LDA 22 W

F-PLN

APPR

STAR

TRANS

RNV22-W

XAC 1B

BACON

CHK IKL, IKL/8 180kts, /3 160kts, HME 015 (GA)



고려사항

- Procedure SPD 지켜야 Traffic 간격 유지됨. 무게에 따라 -4도 증속 가능성 있음.
- S/B 사용 하거나 강하 각을 조절 하는 것이 필요함.

## Missed App

R/H Turn HME 015 OUTBD (PULL TRK)

(DCT KASGA Radial In 196, MAN NAV)

# A330 HND LDA 23 W

F-PLN

APPR

STAR

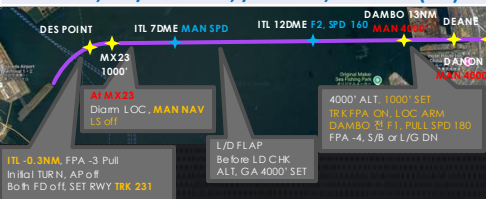
TRANS

RNV23-W

XAC 2B

BACON

CHK ITL, ITL/12 180kts, /7 160kts, HME 177 (GA)



## Missed App

L/H Turn HME 177 OUTBD (PULL TRK)

(DCT UTIBO Radial In 357, MAN NAV)

|                              |                              |
|------------------------------|------------------------------|
| <b><u>RJTT(HND) 21ft</u></b> | <b><u>RKSI(ICN) 23ft</u></b> |
|------------------------------|------------------------------|

|                                |                               |
|--------------------------------|-------------------------------|
| Delta Oper 132.075<br>DCL -15분 | <b><u>PA</u></b> KE ICN 131.5 |
|--------------------------------|-------------------------------|

HND : SID (xx B/C 2200-0230z 0600-1000z) NADP 1

|     |                    |                                                 |
|-----|--------------------|-------------------------------------------------|
| ALL | BEKLA x<br>OPPAR x | HD 338(158), 051(231)<br>ALT ATC<br>Mod SPD 230 |
|-----|--------------------|-------------------------------------------------|

|              |                  |                   |                  |                   |                 |                 |
|--------------|------------------|-------------------|------------------|-------------------|-----------------|-----------------|
| HME<br>112.2 | 34L IHA<br>111.7 | 16R ITA<br>111.55 | 34R ITC<br>108.9 | 16L IOC<br>111.95 | 22 IAD<br>108.1 | 23 ITD<br>110.5 |
|--------------|------------------|-------------------|------------------|-------------------|-----------------|-----------------|

|     |                                                                      |
|-----|----------------------------------------------------------------------|
| FIX | 34L : HME 351/1.1, R095, 34R : HME R080,<br>R095, 22 : HME /2.2 R185 |
|-----|----------------------------------------------------------------------|

34R BEKLA : KAIJI 230kts, TO FLAP SPD Until TORAM  
05 OPPAR : Reduce SPD ARC TURN  
RWY05 RTE5 TAXI Chart



DEP ATIS  
TKO 120.5 – FUK 133.02  
TGU 120.57  
APP 119.75



ICN [333/153] : STAR

|       |          |       |           |
|-------|----------|-------|-----------|
| 33/34 | GUKDO xE | ENPIL | GUKDO 180 |
| 15/16 | GUKDO xH | MUNAN | GUKDO 180 |

|     |                                                                                            |
|-----|--------------------------------------------------------------------------------------------|
| NAV | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR<br>WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |
|-----|--------------------------------------------------------------------------------------------|

|     |                                      |
|-----|--------------------------------------|
| FIX | RWY /8(180kts), /5(160kts), YJU R271 |
|-----|--------------------------------------|

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')  
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')  
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO



|                                                 |                                   |                               |                                   |                    |
|-------------------------------------------------|-----------------------------------|-------------------------------|-----------------------------------|--------------------|
| <u>RKSI(ICN) 23ft</u>                           |                                   | <u>RJFF(FUK) 30ft</u>         |                                   |                    |
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm |                                   | PA                            | KE FUK 132.05                     |                    |
| ICN : SID (33/34 NADP 1, 15/16 NADP 2)          |                                   |                               |                                   |                    |
| 33L/R                                           | OSPOT xE/A                        | HD 333 ALT 5500/ATC SPD 230/x |                                   |                    |
| 34L/R                                           | OSPOT xY                          | HD 333 ALT ATC SPD 230        |                                   |                    |
| 15L/R                                           | OSPOT xC                          | HD 153 ALT 5000 NADP2         |                                   |                    |
| 16L/R                                           | OSPOT xH                          | HD 153 ALT 5000 NADP2         |                                   |                    |
| NCN<br>113.8                                    | 33L INLL<br>109.3                 | 33R INRR<br>108.9             | 15L ISLL<br>111.9                 | 15R ISRR<br>109.1  |
| WNG<br>112.9                                    | 34L IRFN<br>109.95                | 34R IRKN<br>108.1             | 16L IRKS<br>110.35                | 16R IRFS<br>108.55 |
| FIX                                             | 33L/R : NC05L/R, R242<br>YJU R271 |                               | 34L/R : EO34L/R, R242<br>YJU R271 |                    |
| Parallel TWY 10KTS 이상(R17 MAX 15kts)            |                                   |                               |                                   |                    |

[TGU 125.37](#)  
[Kobe 118.9 – FUK APP 119.65](#)  
[FUK RDR – 121.125](#)



FUK [338/158] : RDR Vectoring from IKE  
(PAVGA 13000ft)  
Hold W of IKE published (Confirm FMS DATA)

|     |                                            |                |                            |
|-----|--------------------------------------------|----------------|----------------------------|
| 16  | SARUP                                      | ENTIX          | ILS, RNP, LOC 16           |
| 34  | V34<br>HAWKS WEST                          | RWY34<br>HAWKS | VIS 34<br>ILS, RNP, LOC 34 |
| NAV | DGC, 16 IFO, 34 IFF DGC VOR out of 6NM A/P |                |                            |

16 : C6(5505'), C7(6407'), 34 : C4(5193'), C3(6354')

VIS 34 : After IKE – RDR Vector Downwind – 1800ft –  
RWY Insight 1500ft – Before L/D CHK Complete  
before base (Do not Extend Downwind due Terrain)

**RJFF(FUK) 30ft**

**RKSI(ICN) 23ft**

KE FUK 132.05

DCL -15min, Voice -5min

**PA**

KE ICN 131.5

**FUK : SID (Consider C2, C8 Intersection T/O)**

16

**HAKATA**

**HD 158 ALT ATC (10000)**

34

**XX**

**HD 338 ALT ATC (10000)**

**DGC 114.5**

**16 IFO 111.7**

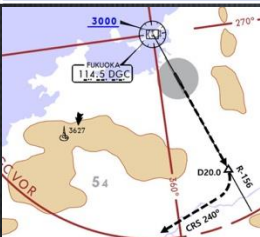
**34 IFF108.9**

**FIX**

16 : DGC 156/20 R240  
(DGC VOR out of 6NM A/P)

**Caution GP HOLD LINE**

Initial CTC TWR, "Ready for departure"  
RWSL(Runway Status Lights) in operation



[DEP 127.9](#)

[Kobe 135.65](#)

[TGU 125.37](#)



**ICN [333/153] : STAR**

**33/34**

**GUkDO xE**

**ENPIL**

**GUkDO 180**

**15/16**

**GUkDO xH**

**MUNAN**

**GUkDO 180**

**NAV**

**NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR**

**WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS**

**FIX**

**RWY /8(180kts), /5(160kts), YJU R271**

**33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')**

**15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')**

**34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')**

**16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')**

**8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO**

|                       |                       |
|-----------------------|-----------------------|
| <b>RKSI(ICN) 23ft</b> | <b>RJGG(NGO) 12ft</b> |
|-----------------------|-----------------------|

|                                                 |                                         |
|-------------------------------------------------|-----------------------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> SWISSPORT<br>OPERATION 132.05 |
|-------------------------------------------------|-----------------------------------------|

|                                        |
|----------------------------------------|
| ICN : SID (33/34 NADP 1, 15/16 NADP 2) |
|----------------------------------------|

|              |                                   |                               |                                  |                    |
|--------------|-----------------------------------|-------------------------------|----------------------------------|--------------------|
| 33L/R        | EGOBA xE/A                        | HD 333 ALT 5500/ATC SPD 230/x |                                  |                    |
| 34L/R        | EGOBA xY                          | HD 333 ALT ATC SPD 230        |                                  |                    |
| 15L/R        | EGOBA xC                          | HD 153 ALT 5000 NADP2         |                                  |                    |
| 16L/R        | EGOBA xH                          | HD 153 ALT 5000 NADP2         |                                  |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9             | 15L ISLL<br>111.9                | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1             | 16L IRKS<br>110.35               | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                               | 34L/R : EO34L/R,R242<br>YJU R271 |                    |

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

|                                                                                                                    |                                                                                      |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <a href="#">DEP 125.15</a><br><a href="#">TGU 134.17 – TKO 133.8 – 133.02</a><br><a href="#">센트레아 APP – 121.05</a> |  |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

|                                             |
|---------------------------------------------|
| NGO [356/176] : STAR (SAMON 290, MARIA 130) |
|---------------------------------------------|

|     |                       |       |          |
|-----|-----------------------|-------|----------|
| 36  | CHESS(CARDS)<br>SOUTH | PROBE | ILS Z 36 |
| 18  | CHESS(CARDS)<br>NORTH | QUEST | ILS Z 18 |
| NAV | CBE, 36 ICX, 18 ICY   |       |          |

|                                      |
|--------------------------------------|
| 36 : A6(5213'), A7(6525'), A8(7837') |
| 18 : A5(5393'), A4(6528'), A3(7841') |

|                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------|
| RWY36 : After 1500ft L/D FLAP<br>RWY 18 : After 3000ft L/G DN & L/D FLAP<br>Caution Stop line, Yellow Ramp line, VDGS!!! |
|--------------------------------------------------------------------------------------------------------------------------|

|                       |                       |
|-----------------------|-----------------------|
| <u>RJGG(NGO) 12ft</u> | <u>RKSI(ICN) 23ft</u> |
|-----------------------|-----------------------|

|                                                                         |                  |              |
|-------------------------------------------------------------------------|------------------|--------------|
| SWISSPORT OPERATION<br>132.05 <span style="color: red;">DCL -15분</span> | <b><u>PA</u></b> | KE ICN 131.5 |
|-------------------------------------------------------------------------|------------------|--------------|

|                               |
|-------------------------------|
| NGO : SID – TANGO tx (NADP 1) |
|-------------------------------|

|    |                      |                      |
|----|----------------------|----------------------|
| 36 | OUMI x<br>- TANGO tx | HD 356 ALT ATC(7000) |
| 18 |                      | HD 176 ALT ATC(7000) |

|           |              |              |
|-----------|--------------|--------------|
| CBE 117.8 | 18 ICY 109.7 | 36 ICX 111.9 |
|-----------|--------------|--------------|

|                                           |
|-------------------------------------------|
| APU Start 30min, Prepare Intersection T/O |
|-------------------------------------------|

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| <a href="#">DEP 120.0</a><br><a href="#">TKO 133.55 – 133.8 – TGU 120.52</a><br><a href="#">APP – 119.75</a> |
|--------------------------------------------------------------------------------------------------------------|



|                      |
|----------------------|
| ICN [333/153] : STAR |
|----------------------|

|       |          |       |           |
|-------|----------|-------|-----------|
| 33/34 | GUkDO xE | ENPIL | GUkDO 180 |
| 15/16 | GUkDO xH | MUNAN | GUkDO 180 |

|     |                                             |
|-----|---------------------------------------------|
| NAV | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR |
|     | WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |

|     |                                      |
|-----|--------------------------------------|
| FIX | RWY /8(180kts), /5(160kts), YJU R271 |
|-----|--------------------------------------|

|                                                        |
|--------------------------------------------------------|
| 33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513') |
| 15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641') |

|                                                        |
|--------------------------------------------------------|
| 34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507') |
| 16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444') |

|                                                  |
|--------------------------------------------------|
| 8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO |
|--------------------------------------------------|

|                       |                       |
|-----------------------|-----------------------|
| <b>RKSI(ICN) 23ft</b> | <b>VHHH(HKG) 28ft</b> |
|-----------------------|-----------------------|

|                                                 |                                     |
|-------------------------------------------------|-------------------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> HAS FLT Dispatch<br>131.6 |
|-------------------------------------------------|-------------------------------------|

|                                        |
|----------------------------------------|
| ICN : SID (33/34 NADP 1, 15/16 NADP 2) |
|----------------------------------------|

|              |                                   |                       |                                   |                    |
|--------------|-----------------------------------|-----------------------|-----------------------------------|--------------------|
| 33L/R        | BOPTA xA                          | HD 333 ALT ATC        |                                   |                    |
| 34L/R        | BOPTA xY                          | HD 333 ALT ATC        |                                   |                    |
| 15L/R        | BOPTA xC                          | HD 153 ALT 5000 NADP2 |                                   |                    |
| 16L/R        | BOPTA xH                          | HD 153 ALT 5000 NADP2 |                                   |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9     | 15L ISLL<br>111.9                 | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1     | 16L IRKS<br>110.35                | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                       | 34L/R : EO34L/R, R242<br>YJU R271 |                    |

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

|                                                                                                                                                                         |                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <a href="#">ICN 124.52(125.72) – FUK 127.5 – TPE 125.5 – 126.7</a><br><a href="#">129.1 – HKG RDR 121.3 – 126.5</a><br><a href="#">DEP 122.0 – Final 119.1 – 119.35</a> |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

|                                                                                                                |
|----------------------------------------------------------------------------------------------------------------|
| <b>HKG [074/254] TL110 : Terminal Tx RTE + STAR Chart</b><br><b>ENPET FL260, RWY25R After TOPUN - APP Mode</b> |
|----------------------------------------------------------------------------------------------------------------|

|              |                          |       |                              |
|--------------|--------------------------|-------|------------------------------|
| 07L/R<br>07C | ABBEY xxA<br>SIERA xxA/C | LIMES | ILS 07L/R/C                  |
| 25R/L<br>25C | ABBEY xxB<br>SIERA xxB/D | RIVMI | RNAV tx ILS 25R<br>ILS 25L/C |

|                                                            |
|------------------------------------------------------------|
| 1500 – 1659 Z RWY 07R/5L no avail, (ALL DIS TH Except 25L) |
|------------------------------------------------------------|

|     |                                                                                 |
|-----|---------------------------------------------------------------------------------|
| NAV | <b>SMT, 07L IZSL, 07R IZSR, 07C IZSC</b><br><b>25R ITFR, 25L ITFL, 25C ITFC</b> |
|-----|---------------------------------------------------------------------------------|

|                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07L : C7(5882'), C8(7194'), 25R : C6(5882'), C5(7211')<br>07R : J7(6916'), J8(7998'), 25L : J5(6916'), J4(8192')<br>07C : A7(5692'), A8(7208'), 25C : A6(5127'), A56673') |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                   |
|---------------------------------------------------------------------------------------------------|
| <b>Confirm Chart SPD vs ATC SPD (ATC First)</b><br>Solid Line for WideBody, APU BAN off Procedure |
|---------------------------------------------------------------------------------------------------|

|                       |                       |
|-----------------------|-----------------------|
| <b>VHHH(HKG) 28ft</b> | <b>RKSI(ICN) 23ft</b> |
|-----------------------|-----------------------|

HAS FLT Disp 131.6

DCL 20분전

5분 차이시 CTC Comm

**PA**

KE ICN 131.5

**HKG : SID + Terminal Tx RTE Chart TA 9000**

**NADP2 : 1000 No LVR CLB - FLAP 0 - LVR CLB**

**3000 Manage SPD (NADP 1/2 for 07L/R/C)**

|                |                                        |                                                               |
|----------------|----------------------------------------|---------------------------------------------------------------|
| <b>07L/C/R</b> | <b>DALOL x E/C/A</b><br>(Z/Y/X RF LEG) | <b>HD 074 ALT 5000 SPD 205</b><br><b>NADP 1/2 (Z/Y/X 210)</b> |
|----------------|----------------------------------------|---------------------------------------------------------------|

|                |                      |                                                         |
|----------------|----------------------|---------------------------------------------------------|
| <b>25L/C/R</b> | <b>DALOL x B/D/F</b> | <b>HD 254 ALT 5000 SPD 205</b><br><b>NADP 2 TA 9000</b> |
|----------------|----------------------|---------------------------------------------------------|

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| <b>NAV</b> | <b>SMT, 07L IZSL, 07R IZSR, 07C IZSC</b><br><b>25R ITFR, 25L ITFL, 25C ITFC</b> |
|------------|---------------------------------------------------------------------------------|

|             |                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E. O</b> | 07L: IZSL074/3, R095, IZSL/9.1, 25R: ITFR254/10, R156<br>07C: IZSC074/2.7, R080, IZSC/6.2, 25C: ITFC254/10, R156<br>07R: IZSR074/6.2, R080, MAX200kt, 25L: ITFL254/6, R145 |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**SID – Tx RTE Chart Many SPD Restriction**



[HKG DEP 123.8 – RDR 118.925](#)

[TPE 129.1 – 126.7 – 123.6 – 125.5](#)

[FUK 127.5 – ICN 125.725\(124.52\)](#)

[ICN – 120.72 – 126.17](#)

[APP – 119.75](#)



**ICN [333/153] : STAR**

|              |                 |              |                  |
|--------------|-----------------|--------------|------------------|
| <b>33/34</b> | <b>OLMEN xE</b> | <b>ENPIL</b> | <b>OLMEN 180</b> |
|--------------|-----------------|--------------|------------------|

|              |                 |              |                  |
|--------------|-----------------|--------------|------------------|
| <b>15/16</b> | <b>OLMEN xH</b> | <b>MUNAN</b> | <b>OLMEN 180</b> |
|--------------|-----------------|--------------|------------------|

|            |                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------|
| <b>NAV</b> | <b>NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR</b><br><b>WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS</b> |
|------------|----------------------------------------------------------------------------------------------------------|

|            |                                             |
|------------|---------------------------------------------|
| <b>FIX</b> | <b>RWY /8(180kts), /5(160kts), YJU R271</b> |
|------------|---------------------------------------------|

**33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')**  
**15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')**

**34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')**  
**16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')**

**8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO**

|                       |                        |
|-----------------------|------------------------|
| <b>RKSI(ICN) 23ft</b> | <b>ZBAA(PEK) 116ft</b> |
|-----------------------|------------------------|

|                                                 |                                      |
|-------------------------------------------------|--------------------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> Air China Beijing<br>132.0 |
|-------------------------------------------------|--------------------------------------|

|                                               |
|-----------------------------------------------|
| <b>ICN : SID (33/34 NADP 1, 15/16 NADP 2)</b> |
|-----------------------------------------------|

|                     |                                   |                              |                                   |                           |
|---------------------|-----------------------------------|------------------------------|-----------------------------------|---------------------------|
| <b>33L/R</b>        | <b>NOPIK xA</b>                   | <b>HD 333 ALT ATC</b>        |                                   |                           |
| <b>34L/R</b>        | NOPIK xY                          | HD 333 ALT ATC               |                                   |                           |
| <b>15L/R</b>        | <b>BINIL xC</b>                   | <b>HD 153 ALT 5000 NADP2</b> |                                   |                           |
| <b>16L/R</b>        | BINIL xH                          | HD 153 ALT 5000 NADP2        |                                   |                           |
| <b>NCN</b><br>113.8 | <b>33L INLL</b><br>109.3          | <b>33R INRR</b><br>108.9     | <b>15L ISLL</b><br>111.9          | <b>15R ISRR</b><br>109.1  |
| <b>WNG</b><br>112.9 | <b>34L IRFN</b><br>109.95         | <b>34R IRKN</b><br>108.1     | <b>16L IRKS</b><br>110.35         | <b>16R IRFS</b><br>108.55 |
| <b>FIX</b>          | 33L/R : NC05L/R, R242<br>YJU R271 |                              | 34L/R : EO34L/R, R242<br>YJU R271 |                           |

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

[DEP 125.15 – TGU 132.8 – DLC 132.95](#)

[TAO 133.72 – 128.15 – PEK 125.6](#)

[PEK APP 120.6 – Final 119.0](#)



|                                                 |
|-------------------------------------------------|
| <b>PEK [001/181] : RW01/19 main (RW36L/18R)</b> |
|-------------------------------------------------|

|                   |                                                                   |              |                         |
|-------------------|-------------------------------------------------------------------|--------------|-------------------------|
| <b>01/36L/36R</b> | <b>DUMAP x Z</b>                                                  | <b>AA141</b> | <b>ILS Z 01 (Y 36L)</b> |
| <b>19/18R/18L</b> | <b>DUMAP x G</b>                                                  | <b>AA261</b> | <b>ILS Z 19 (Y18R)</b>  |
| <b>NAV</b>        | <b>PEK, 01 INJ, 36L IDK, 19 ISZ, 18R ILG<br/>36R IQU, 18L IOR</b> |              |                         |

|                                                        |
|--------------------------------------------------------|
| <b>FIX: RWxx /8(180kts), /6(160kts) TMA Max 280kts</b> |
|--------------------------------------------------------|

|                                                               |
|---------------------------------------------------------------|
| <b>01 : Q5(5223'), Q6(7024'), 19 3.2 :Q4(5298'),Q3(7103')</b> |
| <b>36L : P6(6276'), P7(7719'), 18R : P3(6223'), P2(7552')</b> |

|                                                                                            |
|--------------------------------------------------------------------------------------------|
| <b>APU off Procedure (GND Air Cond' &amp; GPU)<br/>Standard TAXI RTE in Jeppesen Chart</b> |
|--------------------------------------------------------------------------------------------|

# Meter/Feet Conversion Table

Westbound



Eastbound



13,100 m

43,000 ft

12,200 m

40,100 ft

11,600 m

38,100 ft

11,000 m

36,100 ft

10,400 m

34,100 ft

9,800 m

32,100 ft

9,200 m

30,100 ft

8,400 m

27,600 ft

7,800 m

25,600 ft

7,200 m

23,600 ft

6,600 m

21,700 ft

6,000 m

19,700 ft

5,400 m

17,700 ft

4,800 m

15,700 ft

4,200 m

13,800 ft

3,600 m

11,800 ft

3,000 m

9,800 ft

2,400 m

7,900 ft

1,800 m

5,900 ft

1,200 m

3,900 ft

13,700 m

44,900 ft

12,500 m

41,100 ft

11,900 m

39,100 ft

11,300 m

37,100 ft

10,700 m

35,100 ft

10,100 m

33,100 ft

9,500 m

31,100 ft

8,900 m

29,100 ft

8,100 m

26,600 ft

7,500 m

24,600 ft

6,900 m

22,600 ft

6,300 m

20,700 ft

5,700 m

18,700 ft

5,100 m

16,700 ft

4,500 m

14,800 ft

3,900 m

12,800 ft

3,300 m

10,800 ft

2,700 m

8,900 ft

2,100 m

6,900 ft

1,500 m

4,900 ft

## ALT/HEIGHT Conversion

1,000 m

3,300 ft

500 m

1,600 ft

900 m

3,000 ft

450 m

1,500 ft

800 m

2,600 ft

400 m

1,300 ft

700 m

2,300 ft

350 m

1,100 ft

600 m

2,000 ft

300 m

1,000 ft

550 m

1,800 ft



|                        |                       |
|------------------------|-----------------------|
| <b>ZBAA(PEK) 116ft</b> | <b>RKSI(ICN) 23ft</b> |
|------------------------|-----------------------|

|                                                                                  |                        |
|----------------------------------------------------------------------------------|------------------------|
| Air China Beijing 132.0<br>DCL 40분전, Voice 20분전<br>(COBT/STD 15분 차이 CTC<br>Comm) | <b>PA</b> KE ICN 131.5 |
|----------------------------------------------------------------------------------|------------------------|

**PEK : SID (NADP 1) RW36R/18L Intersec T/O W2, W7**

|                    |                         |                                                                  |
|--------------------|-------------------------|------------------------------------------------------------------|
| <b>36R</b><br>(01) | <b>MUGLO x X/Y(x X)</b> | <b>HD 001 ALT 2100m(1800m)</b><br><b>Maintain RWY TRK TA9850</b> |
|--------------------|-------------------------|------------------------------------------------------------------|

|                    |                        |                                                |
|--------------------|------------------------|------------------------------------------------|
| <b>18L</b><br>(19) | <b>MUGLO x G (x J)</b> | <b>HD 181 ALT DCL(1500m)</b><br><b>TA 9850</b> |
|--------------------|------------------------|------------------------------------------------|

|                            |                                 |                                |                               |                               |
|----------------------------|---------------------------------|--------------------------------|-------------------------------|-------------------------------|
| <b>PEK</b><br><b>114.7</b> | <b>36R IQU</b><br><b>111.55</b> | <b>18L IOR</b><br><b>109.3</b> | <b>01 INJ</b><br><b>108.5</b> | <b>19 ISZ</b><br><b>108.9</b> |
|----------------------------|---------------------------------|--------------------------------|-------------------------------|-------------------------------|

|            |                                                                 |
|------------|-----------------------------------------------------------------|
| <b>FIX</b> | 36R : PEK 325/11, 36L : PEK 326/13, 01 : PEK<br>323/9 then R124 |
|------------|-----------------------------------------------------------------|

**COBT from ATIS “Enroute”, Bad Wx DOTRA SID**



[DEP 124.4](#)

[PEK APP 120.6 – PEK 125.6](#)

[DLC 123.2 – 132.95](#)

[ICN 132.8 – APP 119.75](#)



**ICN [333/153] : STAR**

|              |                  |              |                  |
|--------------|------------------|--------------|------------------|
| <b>33/34</b> | <b>REBIT x A</b> | <b>PAMBI</b> | <b>REBIT 170</b> |
|--------------|------------------|--------------|------------------|

|              |                  |              |                  |
|--------------|------------------|--------------|------------------|
| <b>15/16</b> | <b>REBIT x H</b> | <b>MUNAN</b> | <b>REBIT 170</b> |
|--------------|------------------|--------------|------------------|

|            |                                                    |
|------------|----------------------------------------------------|
| <b>NAV</b> | <b>NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR</b> |
|            | <b>WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS</b> |

|            |                                              |
|------------|----------------------------------------------|
| <b>FIX</b> | RWY /8( 180kts), /5(160kts), P518 R068, R278 |
|------------|----------------------------------------------|

**33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')**  
**15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')**

**34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')**  
**16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')**

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, **HIRO**

|                       |                       |
|-----------------------|-----------------------|
| <b>RKSI(ICN) 23ft</b> | <b>ZHHH(WUH)115ft</b> |
|-----------------------|-----------------------|

|                                                  |           |      |
|--------------------------------------------------|-----------|------|
| KE ICN 131.5<br>DCL -10분 TOBT 5분 차이시<br>CTC Comm | <b>PA</b> | None |
|--------------------------------------------------|-----------|------|

**ICN : SID (33/34 NADP 1, 15/16 NADP 2)**

|                     |                                   |                              |                                   |                           |
|---------------------|-----------------------------------|------------------------------|-----------------------------------|---------------------------|
| <b>33L/R</b>        | <b>NOPIK xA</b>                   | <b>HD 333 ALT ATC</b>        |                                   |                           |
| <b>34L/R</b>        | NOPIK xY                          | HD 333 ALT ATC               |                                   |                           |
| <b>15L/R</b>        | <b>BINIL xC</b>                   | <b>HD 153 ALT 5000 NADP2</b> |                                   |                           |
| <b>16L/R</b>        | BINIL xH                          | HD 153 ALT 5000 NADP2        |                                   |                           |
| <b>NCN</b><br>113.8 | <b>33L INLL</b><br>109.3          | <b>33R INRR</b><br>108.9     | <b>15L ISLL</b><br>111.9          | <b>15R ISRR</b><br>109.1  |
| <b>WNG</b><br>112.9 | <b>34L IRFN</b><br>109.95         | <b>34R IRKN</b><br>108.1     | <b>16L IRKS</b><br>110.35         | <b>16R IRFS</b><br>108.55 |
| <b>FIX</b>          | 33L/R : NC05L/R, R242<br>YJU R271 |                              | 34L/R : EO34L/R, R242<br>YJU R271 |                           |

Parallel TWY 10KTS 이상(R17 MAX 15kts)

[DEP 125.15 – TGU 132.8 – DLC 132.95](#)  
[TAO 133.72 – 128.15 – PEK 125.6](#)



**WUH [046/226] : WUH CTL Rep HLxxxx, RDR CTC 60NM**  
**Prepare STAR CHG**

|                 |                                                                          |                    |              |
|-----------------|--------------------------------------------------------------------------|--------------------|--------------|
| <b>04/05R/L</b> | <b>PONUD 01T/05T</b>                                                     | <b>HH310/HH357</b> | <b>ILS Z</b> |
| <b>22/23L/R</b> | <b>PONUD 02T</b>                                                         | <b>AVLEN</b>       | <b>ILS Z</b> |
| <b>NAV</b>      | <b>WHA, 04 IHN, 05R IMU, 22 ITS, 23L IQD,</b><br><b>05L IWF, 23R IUT</b> |                    |              |

FIX: RWxx /8(180kts) from intercept final 180kts

**04 : B8(5557'), B9(6578'), 22 : B5(6738'), B4(7939')**  
**05R : F10(5475'), F11(6597'), 23L : F7(5341'), F6(6545')**

Follow Me Car TWY K/J  
APU off Procedure (GND Air Cond' & GPU)

# Meter/Feet Conversion Table

Westbound



Eastbound



**13,100 m**

**43,000 ft**

12,200 m

40,100 ft

**11,600 m**

**38,100 ft**

11,000 m

36,100 ft

**10,400 m**

**34,100 ft**

9,800 m

32,100 ft

**9,200 m**

**30,100 ft**

8,400 m

27,600 ft

**7,800 m**

**25,600 ft**

7,200 m

23,600 ft

**6,600 m**

**21,700 ft**

6,000 m

19,700 ft

**5,400 m**

**17,700 ft**

4,800 m

15,700 ft

**4,200 m**

**13,800 ft**

**3,600 m**

**11,800 ft**

**3,000 m**

**9,800 ft**

2,400 m

7,900 ft

**1,800 m**

**5,900 ft**

1,200 m

3,900 ft

13,700 m

44,900 ft

**12,500 m**

**41,100 ft**

11,900 m

39,100 ft

**11,300 m**

**37,100 ft**

10,700 m

35,100 ft

**10,100 m**

**33,100 ft**

9,500 m

31,100 ft

**8,900 m**

**29,100 ft**

8,100 m

26,600 ft

**7,500 m**

**24,600 ft**

6,900 m

22,600 ft

**6,300 m**

**20,700 ft**

5,700 m

18,700 ft

**5,100 m**

**16,700 ft**

4,500 m

14,800 ft

**3,900 m**

**12,800 ft**

3,300 m

10,800 ft

**2,700 m**

**8,900 ft**

2,100 m

6,900 ft

**1,500 m**

**4,900 ft**

## ALT/HEIGHT Conversion

1,000 m

3,300 ft

500 m

1,600 ft

**900 m**

**3,000 ft**

**450 m**

**1,500 ft**

800 m

2,600 ft

400 m

1,300 ft

**700 m**

**2,300 ft**

**350 m**

**1,100 ft**

600 m

2,000 ft

300 m

1,000 ft

**550 m**

**1,800 ft**

ZHHH(WUH)115ft RKSI(ICN) 23ft

None  
DCL 10-30분전  
Readback to CLR

**PA**

KE ICN 131.5

WUH : SID (NADP 1) **Readback RWY, ALT to CLR**  
**Report Ready for P/B to CLR**

|              |                 |                                                            |                  |                  |
|--------------|-----------------|------------------------------------------------------------|------------------|------------------|
| 04/<br>05L   | BIVIP 7S        | HD 046 ALT 4800m or<br>ATC(900m) TA 9850                   |                  |                  |
| 22/<br>23R   | BIVIP 6S        | HD 246 ALT 4800m or<br>ATC(900m)<br>SPD 220/230kts TA 9850 |                  |                  |
| WHA<br>112.2 | 04 IHN<br>109.3 | 22 ITS<br>108.5                                            | 05L IWF<br>111.5 | 23R IUT<br>111.5 |

Uncommanded Turn RNP Departure

[DEP 119.575](#)

[SHA 120.55 – 120.95](#)

[ICN 132.8 – APP 119.75](#)



ICN [333/153] : STAR

|       |                                             |       |           |
|-------|---------------------------------------------|-------|-----------|
| 33/34 | OLMEN xE                                    | ENPIL | OLMEN 180 |
| 15/16 | OLMEN xH                                    | MUNAN | OLMEN 180 |
| NAV   | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR |       |           |
|       | WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |       |           |
| FIX   | RWY /8(180kts), /5(160kts), YJU R271        |       |           |

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')  
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')  
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, **HIRO**

|                       |                       |
|-----------------------|-----------------------|
| <b>RKSI(ICN) 23ft</b> | <b>ZGGG(CAN) 50ft</b> |
|-----------------------|-----------------------|

|                                                 |                                   |
|-------------------------------------------------|-----------------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> China Southern<br>132.0 |
|-------------------------------------------------|-----------------------------------|

|                                        |
|----------------------------------------|
| ICN : SID (33/34 NADP 1, 15/16 NADP 2) |
|----------------------------------------|

|              |                                   |                       |                                  |                    |
|--------------|-----------------------------------|-----------------------|----------------------------------|--------------------|
| 33L/R        | BOPTA xA                          | HD 333 ALT ATC        |                                  |                    |
| 34L/R        | BOPTA xY                          | HD 333 ALT ATC        |                                  |                    |
| 15L/R        | BOPTA xC                          | HD 153 ALT 5000 NADP2 |                                  |                    |
| 16L/R        | BOPTA xH                          | HD 153 ALT 5000 NADP2 |                                  |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9     | 15L ISLL<br>111.9                | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1     | 16L IRKS<br>110.35               | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                       | 34L/R : EO34L/R,R242<br>YJU R271 |                    |

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

|                                                                                      |
|--------------------------------------------------------------------------------------|
| <a href="#">DEP 125.15 -</a>                                                         |
|  |

|                                                                                               |
|-----------------------------------------------------------------------------------------------|
| CAN [017/197] : CAN CTL, APP Freq no Chart<br>Short PTN AFT IAF, RW20 Windshear, Lower GA ALT |
|-----------------------------------------------------------------------------------------------|

|             |                                                                                |             |       |
|-------------|--------------------------------------------------------------------------------|-------------|-------|
| 02R/L,01L/R | OLPAB xx                                                                       | GG611/GG612 | ILS Z |
| 20L/R,19R/L |                                                                                | GG511/GG513 | ILS Z |
| NAV         | POU, 02R IDM, 01L IGW, 20L IXL, 19R<br>IKU, 02L IBB, 01R IOO, 20R IAA, 19L IPP |             |       |

|                                                |
|------------------------------------------------|
| Limit : RWxx /8(180kts) from final, /6(160kts) |
|------------------------------------------------|

|                                                                                                         |
|---------------------------------------------------------------------------------------------------------|
| 02R:Y10(6227'),Y8(7263'), 20L:Y12(6217'),Y14(7240')<br>01L:N5(6453'),N4(7755'), 19R:N8(6423'),N9(7742') |
|---------------------------------------------------------------------------------------------------------|

|                                                                                           |
|-------------------------------------------------------------------------------------------|
| STD Taxi RTE in JEPP(GREEN x), EAT 3/4 in JEPP<br>APU off Procedure (GND Air Cond' & GPU) |
|-------------------------------------------------------------------------------------------|

# Meter/Feet Conversion Table

Westbound



Eastbound



**13,100 m**

**43,000 ft**

12,200 m

40,100 ft

**11,600 m**

**38,100 ft**

11,000 m

36,100 ft

**10,400 m**

**34,100 ft**

9,800 m

32,100 ft

**9,200 m**

**30,100 ft**

8,400 m

27,600 ft

**7,800 m**

**25,600 ft**

7,200 m

23,600 ft

**6,600 m**

**21,700 ft**

6,000 m

19,700 ft

**5,400 m**

**17,700 ft**

4,800 m

15,700 ft

**4,200 m**

**13,800 ft**

**3,600 m**

**11,800 ft**

**3,000 m**

**9,800 ft**

2,400 m

7,900 ft

**1,800 m**

**5,900 ft**

1,200 m

3,900 ft

13,700 m

44,900 ft

**12,500 m**

**41,100 ft**

11,900 m

39,100 ft

**11,300 m**

**37,100 ft**

10,700 m

35,100 ft

**10,100 m**

**33,100 ft**

9,500 m

31,100 ft

**8,900 m**

**29,100 ft**

8,100 m

26,600 ft

**7,500 m**

**24,600 ft**

6,900 m

22,600 ft

**6,300 m**

**20,700 ft**

5,700 m

18,700 ft

**5,100 m**

**16,700 ft**

4,500 m

14,800 ft

**3,900 m**

**12,800 ft**

3,300 m

10,800 ft

**2,700 m**

**8,900 ft**

2,100 m

6,900 ft

**1,500 m**

**4,900 ft**

## ALT/HEIGHT Conversion

1,000 m

3,300 ft

500 m

1,600 ft

**900 m**

**3,000 ft**

**450 m**

**1,500 ft**

800 m

2,600 ft

400 m

1,300 ft

**700 m**

**2,300 ft**

**350 m**

**1,100 ft**

600 m

2,000 ft

300 m

1,000 ft

**550 m**

**1,800 ft**

ZHHH(CAN) 50ft

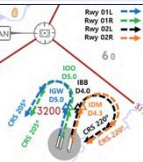
RKSI(ICN) 23ft

|                                                   |                        |
|---------------------------------------------------|------------------------|
| China Southem 132.0<br>DCL 10-30분전<br>No Readback | <b>PA</b> KE ICN 131.5 |
|---------------------------------------------------|------------------------|

CAN : SID (NADP 1) Extra Fuel Delay/CRZ LVL

|                          |                                                          |                                                  |                   |                   |  |
|--------------------------|----------------------------------------------------------|--------------------------------------------------|-------------------|-------------------|--|
| 02L/R,<br>01R/L          | BOVMA<br>xx<br>RNAV                                      | HD 017 ALT 8900m/ATC(900m)<br>SPD 230kts TA 9850 |                   |                   |  |
| 20R/L,<br>19L/R          |                                                          | HD 246 ATC(900m)<br>SPD 230kts TA 9850           |                   |                   |  |
| POU<br>114.1<br>(no APT) | 02R IDM<br>108.5                                         | 01L IGW<br>109.5                                 | 20L IXL<br>111.9  | 19R IKU<br>108.95 |  |
|                          | 02L IBB<br>110.35                                        | 01R IOO<br>109.3                                 | 20R IAA<br>110.75 | 19L IPP<br>111.5  |  |
| FIX                      | 01L/R : IGW/IOO D5.0 R205, 02L/R : IBB/IDM D4.0/4.3 R220 |                                                  |                   |                   |  |

STD Taxi RTE in JEPP(YELLOW x), Report RWY on DEP



DEP 119.7 -



ICN [333/153] : STAR

|       |                                             |       |           |
|-------|---------------------------------------------|-------|-----------|
| 33/34 | OLMEN xE                                    | ENPIL | OLMEN 180 |
| 15/16 | OLMEN xH                                    | MUNAN | OLMEN 180 |
| NAV   | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR |       |           |
|       | WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |       |           |
| FIX   | RWY /8(180kts), /5(160kts) , YJU R271       |       |           |

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')  
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')  
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

|                       |                                |
|-----------------------|--------------------------------|
| <b>RKSI(ICN) 23ft</b> | <b>ZLXY(XIY) <u>1572ft</u></b> |
|-----------------------|--------------------------------|

|                                                 |           |                                   |
|-------------------------------------------------|-----------|-----------------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> | Airport Operation<br>Center 132.0 |
|-------------------------------------------------|-----------|-----------------------------------|

**ICN : SID (33/34 NADP 1, 15/16 NADP 2)**

|                     |                                   |                              |                                   |                           |
|---------------------|-----------------------------------|------------------------------|-----------------------------------|---------------------------|
| <b>33L/R</b>        | <b>NOPIK xA</b>                   | <b>HD 333 ALT ATC</b>        |                                   |                           |
| <b>34L/R</b>        | NOPIK xY                          | HD 333 ALT ATC               |                                   |                           |
| <b>15L/R</b>        | <b>BINIL xC</b>                   | <b>HD 153 ALT 5000 NADP2</b> |                                   |                           |
| <b>16L/R</b>        | BINIL xH                          | HD 153 ALT 5000 NADP2        |                                   |                           |
| <b>NCN</b><br>113.8 | <b>33L INLL</b><br>109.3          | <b>33R INRR</b><br>108.9     | <b>15L ISLL</b><br>111.9          | <b>15R ISRR</b><br>109.1  |
| <b>WNG</b><br>112.9 | <b>34L IRFN</b><br>109.95         | <b>34R IRKN</b><br>108.1     | <b>16L IRKS</b><br>110.35         | <b>16R IRFS</b><br>108.55 |
| <b>FIX</b>          | 33L/R : NC05L/R, R242<br>YJU R271 |                              | 34L/R : EO34L/R, R242<br>YJU R271 |                           |

Parallel TWY 10KTS 이상(R17 MAX 15kts)

|                                                                                      |
|--------------------------------------------------------------------------------------|
| <a href="#">DEP 125.15 – TGU 132.8 – DLC 132.95</a>                                  |
| <a href="#">TAO 133.725 – 128.15</a>                                                 |
| <a href="#">PEK 125.6 – 120.35 – 133.65 – 134.15 – 126.7</a>                         |
| <a href="#">XIY 125.3 – 120.95</a>                                                   |
| <a href="#">XIY APP 119.05 – 120.2 – 125.1</a>                                       |
|  |

**XIY [052/232] : STAR TL118**  
**Spd Restriction at REF Page**  
**Req ILS APP instead of Visual APP**

|              |                                                |              |                         |
|--------------|------------------------------------------------|--------------|-------------------------|
| <b>05L/R</b> | <b>LOVRA xx W</b>                              | <b>XY906</b> | <b>RNAV ILS Z 05L/R</b> |
| <b>23R/L</b> | <b>LOVRA xx Y</b>                              | <b>XY801</b> | <b>RNAV ILS Z 23R/L</b> |
| <b>NAV</b>   | <b>LCZ, 05L IQU, 23R IRA, 05R IXW, 23L IAQ</b> |              |                         |

**05L : A3(6778'), A2(9032'), 23R : A6(5544'), A7(6512')**  
**05R : D4(5613'), D3(7322'), 23L : D5(5646'), D6(7408')**

Follow Me Car, CTC Apron before Gate in  
 “Closing to xx TWY, apply to change to xx Freq”  
 Taxi RTE in Jeppesen Chart.



# Meter/Feet Conversion Table

Westbound



Eastbound



**13,100 m**

**43,000 ft**

12,200 m

40,100 ft

**11,600 m**

**38,100 ft**

11,000 m

36,100 ft

**10,400 m**

**34,100 ft**

9,800 m

32,100 ft

**9,200 m**

**30,100 ft**

8,400 m

27,600 ft

**7,800 m**

**25,600 ft**

7,200 m

23,600 ft

**6,600 m**

**21,700 ft**

6,000 m

19,700 ft

**5,400 m**

**17,700 ft**

4,800 m

15,700 ft

**4,200 m**

**13,800 ft**

**3,600 m**

**11,800 ft**

**3,000 m**

**9,800 ft**

2,400 m

7,900 ft

**1,800 m**

**5,900 ft**

1,200 m

3,900 ft

13,700 m

44,900 ft

**12,500 m**

**41,100 ft**

11,900 m

39,100 ft

**11,300 m**

**37,100 ft**

10,700 m

35,100 ft

**10,100 m**

**33,100 ft**

9,500 m

31,100 ft

**8,900 m**

**29,100 ft**

8,100 m

26,600 ft

**7,500 m**

**24,600 ft**

6,900 m

22,600 ft

**6,300 m**

**20,700 ft**

5,700 m

18,700 ft

**5,100 m**

**16,700 ft**

4,500 m

14,800 ft

**3,900 m**

**12,800 ft**

3,300 m

10,800 ft

**2,700 m**

**8,900 ft**

2,100 m

6,900 ft

**1,500 m**

**4,900 ft**

## ALT/HEIGHT Conversion

1,000 m

3,300 ft

500 m

1,600 ft

**900 m**

**3,000 ft**

**450 m**

**1,500 ft**

800 m

2,600 ft

400 m

1,300 ft

**700 m**

**2,300 ft**

**350 m**

**1,100 ft**

600 m

2,000 ft

300 m

1,000 ft

**550 m**

**1,800 ft**

|                                |                              |
|--------------------------------|------------------------------|
| <u>ZLXY(XIY)</u> <b>1572ft</b> | <u>RKSI(ICN)</u> <b>23ft</b> |
|--------------------------------|------------------------------|

|                                                                |                        |
|----------------------------------------------------------------|------------------------|
| Airport Operation Center<br>132.0<br>DCL -30~10 Min, Read Back | <b>PA</b> KE ICN 131.5 |
|----------------------------------------------------------------|------------------------|

XIY (TA 9850') : RNAV SID (NADP 1)

|           |            |                                        |
|-----------|------------|----------------------------------------|
| 05L<br>/R | WJC xx W/Z | HD 052 ALT ATC<br>1500m(4900') TA 9850 |
|-----------|------------|----------------------------------------|

|           |            |                                            |
|-----------|------------|--------------------------------------------|
| 23R<br>/L | WJC xx X/Y | HD 232 ALT ATC<br>1500m(4900') SPD 205/230 |
|-----------|------------|--------------------------------------------|

|           |                   |                  |                  |                  |
|-----------|-------------------|------------------|------------------|------------------|
| LCZ 109.0 | 05L IQU<br>108.55 | 23R IRA<br>111.7 | 05R IXW<br>109.3 | 23L IAQ<br>111.1 |
|-----------|-------------------|------------------|------------------|------------------|

|     |                 |
|-----|-----------------|
| FIX | 23R/L : LCZ /18 |
|-----|-----------------|

NOTAM TO Perf, ADT = CTOT



[DEP 119.9- XIY 120.95 - 124.1](#)  
[PEK 126.7 - 134.15 - 128.3 - 120.35](#)  
[DLC 123.2 - 132.95](#)  
[TAE 132.8](#)



ICN [333/153] : STAR

|       |           |       |           |
|-------|-----------|-------|-----------|
| 33/34 | REBIT x A | PAMBI | REBIT 170 |
| 15/16 | REBIT x H | MUNAN | REBIT 170 |

|     |                                                                                            |
|-----|--------------------------------------------------------------------------------------------|
| NAV | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR<br>WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |
|-----|--------------------------------------------------------------------------------------------|

|     |                                             |
|-----|---------------------------------------------|
| FIX | RWY /8(180kts), /5(160kts), P518 R068, R278 |
|-----|---------------------------------------------|

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')  
 15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')  
 16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO

| <u>RKPK(PUS) 13ft</u>                                                            |                     | <u>VTBS(BKK) 4ft</u>                 |                                        |
|----------------------------------------------------------------------------------|---------------------|--------------------------------------|----------------------------------------|
| KE Gimhae 129.2<br>DCL -5분                                                       |                     | PA                                   | KE Bangkok<br>131.25                   |
| PUS : NADP CLB 1500, 360000lbs TOGA, CONF 1+F                                    |                     |                                      |                                        |
| 36                                                                               | SOORO x<br>TOPAX tx | HD 002 ALT ATC Modi NADP             |                                        |
| 18                                                                               | BULIM x<br>ENGOT tx | HD 182 ALT 5000 SPD 230<br>Modi NADP |                                        |
| KMH 113.8                                                                        |                     | PSN 114.0                            | 36L IKMA<br>108.5<br>36R IKHE<br>109.5 |
| HUD                                                                              |                     | 36 : KMH R091, R271, R185            |                                        |
| RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS                                     |                     |                                      |                                        |
|  |                     |                                      |                                        |

[DEP 125.5 – TGU 128.17 – 124.52\(125.72\)](#)  
[FUK 127.5\(SENKA /20\)](#)  
[TPE 125.5 – 129.1 – HKG 132.15 – 127.1](#)  
[SNY 122.6 – HNI 123.3 – VTN 128.3](#)  
[BKK 132.1 – 133.1 – APP 119.1](#)



|                                                                                                                                                                            |                                           |           |                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------|--------------------------|
| BKK [015/295] : TL130 UTC+7 (SPD CTL via STAR Chart)                                                                                                                       |                                           |           |                          |
| 19,20R/L                                                                                                                                                                   | EASTE xxC                                 | Select Tx | ILS Z 19/20L<br>RNP 20R  |
| 01,02L/R                                                                                                                                                                   | EASTE xxD                                 | Select Tx | ILS Z 010/02R<br>RNP 02L |
| NAV                                                                                                                                                                        | SVB, 01 ISES, 02R ISWS, 19 ISEN, 20L ISWN |           |                          |
| FIX                                                                                                                                                                        | RWxx /8 (180tks), /5 (160-150kts)         |           |                          |
| 19 : B8(5567'), B10(6965'), 01 : B7(5964'), B5(7962')<br>20L : E9(5052'), E13(7139'), 02R : E12(4872'), E7(6958')<br>20R : F7(6994'), F9(8471'), 02L: F6(5725'), F4(7496') |                                           |           |                          |
| HIRO, 01/19, 02L/20R No Groov, AFT LD CONF 1<br>Standard Taxi Route, APU Off Procedure                                                                                     |                                           |           |                          |

**VTBS(BKK) 4ft**

**RKPK(PUS) 13ft**

KE Bangkok 131.25  
DCL -20min, Voice 133.8

**PA**

KE Gimhae  
129.2

**BKK : RNAV SID (NADP 1) TA 11000**  
**A-CDM REQ Pushback +-5min of TSAT**  
**TSAT/CTOT Inform to GND CTL**

**19,20L/  
R**

**LIPLI x J, G/E**

**HD 195 ALT 6000 19 SPD 220**  
**TA 11000**

**01,02R/  
L**

**LIPLI x K, F/H**

**HD 015 ALT 6000 01 SPD 220**  
**TA 11000**

**SVB**  
**111.4**

**19 ISEN**  
**110.5**

**02R ISWS**  
**109.1**

**20L ISWN**  
**109.5**

**01 ISES**  
**110.1**

APU Start within 10min, Standard TAXI Route  
**19R Do not Pass E1, D2**  
**01/19, 02L/20R No Groov**

DEP 119.25 (AUTO) – BKK 133.1 – VIE 128.3

HNI 123.3 – SNY 122.6 – HKG 127.1 – 125.35

TPE 129.1(126.7, 127.9) – 125.5

FUK 127.5 (SENKA /20)

ICN 125.725(124.52) – 128.17

APP 125.5



**PUS [002/182] T/W 10tks, 36R, 360000lbs 제한**

**ILS 36**

**KEVOX x**

**MASTA**

**9DME LG, 8DME FLAP**

**VOR 18**

**GAYHA x**

**MASTA**

**18 Circling Click!!**

**NAV**

**KMH KMH, 36L IKMA, 36R IKHE**

**FIX**

**36 : IKMA/IKHE /9, /8**

**18 : KMH R284, R280**

**36L : C4 (6299'), C2(7795') / 36R : E3(5866'), E2(7339')**  
**18R(DIS): C6(5770'), C7(6824') 18L:E4(5882'), E5(8792')**

Vacate **C3, C4** by ATC only, Max Taxi SPD 20KTS  
C2 HOLD SHORT 가까움(Vacate TaxiSPD)

| RKSI(ICN) 23ft                                  | VTBS(BKK) 4ft        |
|-------------------------------------------------|----------------------|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | KE Bangkok<br>131.25 |

PA

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

|              |                                   |                       |                                   |                    |
|--------------|-----------------------------------|-----------------------|-----------------------------------|--------------------|
| 33L/R        | BOPTA xA                          | HD 333 ALT ATC        |                                   |                    |
| 34L/R        | BOPTA xY                          | HD 333 ALT ATC        |                                   |                    |
| 15L/R        | BOPTA xC                          | HD 153 ALT 5000 NADP2 |                                   |                    |
| 16L/R        | BOPTA xH                          | HD 153 ALT 5000 NADP2 |                                   |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9     | 15L ISLL<br>111.9                 | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1     | 16L IRKS<br>110.35                | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                       | 34L/R : EO34L/R, R242<br>YJU R271 |                    |

Parallel TWY 10KTS 이상(R17 MAX 15kts)

[FUK 127.5](#)([SENKA /20](#))  
[TPE 125.5 – 129.1 – HKG 132.15 – 127.1](#)  
[SNY 122.6 – HNI 123.3 – VTN 128.3](#)  
[BKK 132.1 – 133.1 – APP 119.1](#)



BKK [015/295] : TL130 UTC+7 (SPD CTL via STAR Chart)

|          |                                           |                      |                          |
|----------|-------------------------------------------|----------------------|--------------------------|
| 19,20R/L | EASTE xxC                                 | Select Tx<br>(No Tx) | ILS Z 19/20L<br>RNP 20R  |
| 01,02L/R | EASTE xxD                                 | Select Tx<br>(No Tx) | ILS Z 010/02R<br>RNP 02L |
| NAV      | SVB, 01 ISES, 02R ISWS, 19 ISEN, 20L ISWN |                      |                          |
| FIX      | RWxx /8 (180tks), /5 (160-150kts)         |                      |                          |

19 : B8(5567'), B10(6965'), 01 : B7(5964'), B5(7962')  
20L : E9(5052'), E13(7139'), 02R : E12(4872'), E7(6958')  
20R : F7(6994'), F9(8471'), 02L : F6(5725'), F4(7496')

HIRO, 01/19, 02L/20R No Groov, AFT LD CONF 1  
Standard Taxi Route, APU Off Procedure

|                      |                       |
|----------------------|-----------------------|
| <u>VTBS(BKK) 4ft</u> | <u>RKSI(ICN) 23ft</u> |
|----------------------|-----------------------|

|                                              |           |              |
|----------------------------------------------|-----------|--------------|
| KE Bangkok 131.25<br>DCL -20min, Voice 133.8 | <b>PA</b> | KE ICN 131.5 |
|----------------------------------------------|-----------|--------------|

|                                                                                                      |
|------------------------------------------------------------------------------------------------------|
| BKK : RNAV SID (NADP 1) TA 11000<br>A-CDM REQ Pushback +-5min of TSAT<br>TSAT/CTOT Inform to GND CTL |
|------------------------------------------------------------------------------------------------------|

|              |                  |                                        |                   |                  |
|--------------|------------------|----------------------------------------|-------------------|------------------|
| 19,20L/<br>R | LIPLI x J, G/E   | HD 195 ALT 6000 19 SPD 220<br>TA 11000 |                   |                  |
| 01,02R/<br>L | LIPLI x K, F/H   | HD 015 ALT 6000 01 SPD 220<br>TA 11000 |                   |                  |
| SVB<br>111.4 | 19 ISEN<br>110.5 | 02R ISWS<br>109.1                      | 20L ISWN<br>109.5 | 01 ISES<br>110.1 |

|                                                                                                  |
|--------------------------------------------------------------------------------------------------|
| APU Start within 10min, Standard TAXI Route<br>19R Do not Pass E1, D2<br>01/19, 02L/20R No Groov |
|--------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                              |                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| DEP 119.25 (AUTO) – BKK 133.1 – VIE 128.3<br>HNI 123.3 – SNY 122.6 – HKG 127.1 – 125.35<br>TPE 129.1(126.7, 127.9) – 125.5<br>FUK 127.5 (SENKA /20)<br>ICN 125.725(124.52) – 120.72 – 126.17 |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

|                                                        |                                             |       |           |
|--------------------------------------------------------|---------------------------------------------|-------|-----------|
| ICN [333/153] : STAR                                   |                                             |       |           |
| 33/34                                                  | OLMEN xE                                    | ENPIL | OLMEN 180 |
| 15/16                                                  | OLMEN xH                                    | MUNAN | OLMEN 180 |
| NAV                                                    | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR |       |           |
|                                                        | WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |       |           |
| FIX                                                    | RWY /8(180kts), /5(160kts) , YJU R271       |       |           |
| 33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513') |                                             |       |           |
| 15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641') |                                             |       |           |
| 34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507') |                                             |       |           |
| 16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444') |                                             |       |           |
| 8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO        |                                             |       |           |

|                                                 |                                   |
|-------------------------------------------------|-----------------------------------|
| <b><u>RKSI(ICN) 23ft</u></b>                    | <b><u>WSSS(SIN) 22ft</u></b>      |
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> SATS Flt Oper<br>131.22 |

**ICN : SID (33/34 NADP 1, 15/16 NADP 2)**

|                     |                                   |                              |                                   |                           |
|---------------------|-----------------------------------|------------------------------|-----------------------------------|---------------------------|
| <b>33L/R</b>        | <b>BOPTA xA</b>                   | <b>HD 333 ALT ATC</b>        |                                   |                           |
| <b>34L/R</b>        | BOPTA xY                          | HD 333 ALT ATC               |                                   |                           |
| <b>15L/R</b>        | <b>BOPTA xC</b>                   | <b>HD 153 ALT 5000 NADP2</b> |                                   |                           |
| <b>16L/R</b>        | BOPTA xH                          | HD 153 ALT 5000 NADP2        |                                   |                           |
| <b>NCN</b><br>113.8 | <b>33L INLL</b><br>109.3          | <b>33R INRR</b><br>108.9     | <b>15L ISLL</b><br>111.9          | <b>15R ISRR</b><br>109.1  |
| <b>WNG</b><br>112.9 | <b>34L IRFN</b><br>109.95         | <b>34R IRKN</b><br>108.1     | <b>16L IRKS</b><br>110.35         | <b>16R IRFS</b><br>108.55 |
| <b>FIX</b>          | 33L/R : NC05L/R, R242<br>YJU R271 |                              | 34L/R : EO34L/R, R242<br>YJU R271 |                           |

Parallel TWY 10KTS 이상(R17 MAX 15kts)

SENKA /20 ETA KABAM 119.3 – ETA MIGUG  
MAVRA 8942/5655 – MIGUG Rep MESOX 120.7  
Rep MAPNO  
(RPHI – VVHM CPDLC Tx Manual)



**SIN [023/203] TL130 UTC+8, CPDLC WSJC -10min FIR**  
**SPD CTL via STAR Chart, No IAF RDR Vector**

|                                                               |                                                        |              |                  |
|---------------------------------------------------------------|--------------------------------------------------------|--------------|------------------|
| <b>02L/C</b>                                                  | <b>MABAL/KARTO xxA</b>                                 | <b>No Tx</b> | <b>ILS 02L/C</b> |
| <b>20R/C</b>                                                  | <b>MABAL/KARTO xxB</b>                                 | <b>No Tx</b> | <b>ILS 20R/C</b> |
| <b>RNP 02R/20L : L/D CHK -&gt; LUVUL/OBGIS -&gt; Arm APPR</b> |                                                        |              |                  |
| <b>NAV</b>                                                    | <b>VTK 02L ICW, 02C ICE, 02R ICX, 20R ICH, 20C ICC</b> |              |                  |
| <b>FIX</b>                                                    | <b>RWxx /8 (180tkts), /4 (150kts)</b>                  |              |                  |

02L : W5(6519'), W4(7916'), 20R : W6(5498'), W7(6423')  
 02C : T6(6781'), T5 (8467'), 20C : T7 (6420'), T8 (7910')  
 02R : A6(6738'), A5 (7995'), 20L : A7 (6250'), A8 (7395')

**HIRO Unable ATC, Exit TWY 2 way but Same Name**  
**VTK Out of A/P, TWY R/S Max 20kts, AFT LD CONF 1**

WSSS(SIN) 22ft

RKSI(ICN) 23ft

SATS Flt Oper 131.22

DCL -20min or Voice

**PA**

KE ICN 131.5

SIN : RNAV SID (NADP 1) TA 11000

REQ P/B +5min of TSAT, CPDLC WSJC Before DEP

20C/R  
20L

TOMAN xx  
CHANGI xD

HD 203 ALT 3000 SPD 230  
230kts Until 4000 TA11000

02C/L/R

TOMAN xx  
CHANGI xC

HD 023 ALT 3000 SPD 230  
230kts Until 4000 TA11000

VTK  
116.5

20C ICC  
109.7

20R ICH  
108.9

02C ICE  
108.3

02L ICW  
110.9

G18-21 Standard P/B, TWY R/S Max 20kts

Cross LUSMO xxx, T/O Before xxx

RWY 20C/R : CHK Min Climb Grad by NAVBLUE

DEP 120.3

ARESI 8942/5655 Unable AKOTA 119.3

SENKA /20



ICN [333/153] : STAR

33/34

OLMEN xE

ENPIL

OLMEN 180

15/16

OLMEN xH

MUNAN

OLMEN 180

NAV

NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR

WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS

FIX

RWY /8(180kts), /5(160kts), YJU R271

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')

15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')

16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, HIRO



|                       |                       |
|-----------------------|-----------------------|
| <b>RKSI(ICN) 23ft</b> | <b>VVDN(DAD) 30ft</b> |
|-----------------------|-----------------------|

|                                                 |           |      |
|-------------------------------------------------|-----------|------|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> | None |
|-------------------------------------------------|-----------|------|

|                                        |
|----------------------------------------|
| ICN : SID (33/34 NADP 1, 15/16 NADP 2) |
|----------------------------------------|

|              |                                   |                       |                                  |                    |
|--------------|-----------------------------------|-----------------------|----------------------------------|--------------------|
| 33L/R        | BOPTA xA                          | HD 333 ALT ATC        |                                  |                    |
| 34L/R        | BOPTA xY                          | HD 333 ALT ATC        |                                  |                    |
| 15L/R        | BOPTA xC                          | HD 153 ALT 5000 NADP2 |                                  |                    |
| 16L/R        | BOPTA xH                          | HD 153 ALT 5000 NADP2 |                                  |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9     | 15L ISLL<br>111.9                | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1     | 16L IRKS<br>110.35               | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                       | 34L/R : EO34L/R,R242<br>YJU R271 |                    |

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

|                                                                                              |                                                                                      |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <a href="#">FUK 127.5</a> ( <a href="#">SENKA /20</a> )<br><a href="#">TPE 125.5 – 129.1</a> |  |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

|                                                                                                    |
|----------------------------------------------------------------------------------------------------|
| DAD [353/173] : STAR TL100 UTC+7, C/S xxx Heavy<br>SPD CTL via STAR Chart, Do Not VOR 17 CAAV Auth |
|----------------------------------------------------------------------------------------------------|

|       |                                    |        |               |
|-------|------------------------------------|--------|---------------|
| 35L/R | BUNTA x M/N/L/K<br>L/K : RNAV Arrs | TAMLA  | ILS V 35L/R   |
| 17R/L | ILS Y 35R -> Circle                | Vector | VIS 17R/L 3.5 |

|     |                                |
|-----|--------------------------------|
| NAV | DAN, 35L DAD, 35R IDR          |
| FIX | RWxx /12 (200tks), /5 (160kts) |

|                                                          |
|----------------------------------------------------------|
| 35L : G4(6702'), G6(9803'), 17R : G2(7089'), G1(9776')   |
| 35R : E4(7198'), E6(10275'), 17L : E2(8044'), E1(10754') |
| NO RAPID EXIT TWY, 90 Degree TURN, AFT LD CONF 1         |

|                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------|
| Follow me Car on E, E2,3,4 Narrow Over-steering<br>TWY D 사용 금지, 소형기 on D시만 E 사용가능<br>Gate IN From TWY E Only, 5m 1kts Ready for STOP |
|--------------------------------------------------------------------------------------------------------------------------------------|

|                       |                       |
|-----------------------|-----------------------|
| <b>VVDN(DAD) 30ft</b> | <b>RKSI(ICN) 23ft</b> |
|-----------------------|-----------------------|

|                             |           |              |
|-----------------------------|-----------|--------------|
| None<br>DCL -20min or Voice | <b>PA</b> | KE ICN 131.5 |
|-----------------------------|-----------|--------------|

**DAD : RNAV SID (NADP 1) TA 9000**  
**Consider Extra Fuel Due FL**

|              |                                |                                                |
|--------------|--------------------------------|------------------------------------------------|
| <b>35R/L</b> | <b>BUNTA xG/C</b><br>RNAV Deps | <b>HD 353 ALT 8000/ATC</b><br><b>TA9000</b>    |
| <b>17L/R</b> | <b>BUNTA xA</b>                | <b>HD 173 ALT ATC SPD 240</b><br><b>TA9000</b> |

|                  |                      |                      |
|------------------|----------------------|----------------------|
| <b>DAN 114.4</b> | <b>35R IDR 111.5</b> | <b>35L DAD 110.5</b> |
|------------------|----------------------|----------------------|

|            |                                                      |
|------------|------------------------------------------------------|
| <b>FIX</b> | 35L/R : DAN 353/5, R060<br>17L/R : DAN 173/4.5, R130 |
|------------|------------------------------------------------------|

Complete P/B – ENG Start, Follow me Car  
**35R CAT I Hold Point Before E1, 17L Hold Short** 가까움  
**TWY D 사용 금지, 소형기 on D시만 E 사용가능**



[DEP 120.45 – DAD](#)

[FUK 127.5 \(SENKA /20\)](#)

[ICN 125.725\(124.52\)](#)

[- 120.72 – 126.17](#)



**ICN [333/153] : STAR**

|              |                                                                                                          |              |                  |
|--------------|----------------------------------------------------------------------------------------------------------|--------------|------------------|
| <b>33/34</b> | <b>OLMEN xE</b>                                                                                          | <b>ENPIL</b> | <b>OLMEN 180</b> |
| <b>15/16</b> | <b>OLMEN xH</b>                                                                                          | <b>MUNAN</b> | <b>OLMEN 180</b> |
| <b>NAV</b>   | <b>NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR</b><br><b>WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS</b> |              |                  |
| <b>FIX</b>   | RWY /8(180kts), /5(160kts), YJU R271                                                                     |              |                  |

**33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')**  
**15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')**  
**34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')**  
**16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')**

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, **HIRO**



|                                    |                        |
|------------------------------------|------------------------|
| <b>VVTS(SGN) 33ft</b>              | <b>RKSI(ICN) 23ft</b>  |
| None<br>-15min, DEL 121.8 By Voice | <b>PA</b> KE ICN 131.5 |

**SGN : RNP SID (NADP 1) TA 1800'**  
**Request RWY due to Performance, Sudden M2000ft**

|           |                  |                                            |                  |
|-----------|------------------|--------------------------------------------|------------------|
| 25L(R)    | KADUM xxD/F      | HD 250 ALT 11000/ATC<br>SPD x/240 TA 18000 |                  |
| 07L(R)    | KADUM xxA        | HD 070 ALT ATC TA 18000                    |                  |
| TSH 116.8 | 25R HCM<br>110.5 | 07R ITS 111.7                              | 25L SGN<br>108.3 |

**Caution TSAT +- 5min, ATC CLR, RWY CHG After TAXI**  
**Caution Sudden STOPBAR L/T, Follow Car Service**  
**1,2,3T is Stand name, Caution S1, P1 for Lineup**

APP 125.5 – HCM 120.1 – 134.05  
HNI 123.3 – SNY 122.6(-5min)  
HKG 132.15 – 127.1 – TPE 129.1 – 127.9  
126.7 – 123.6 – FUK 127.5(SENKA /20)



| ICN [333/153] : STAR                                   |                                             |       |           |
|--------------------------------------------------------|---------------------------------------------|-------|-----------|
| 33/34                                                  | OLMEN xE                                    | ENPIL | OLMEN 180 |
| 15/16                                                  | OLMEN xH                                    | MUNAN | OLMEN 180 |
| NAV                                                    | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR |       |           |
|                                                        | WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |       |           |
| FIX                                                    | RWY /8(180kts), /5(160kts) , YJU R271       |       |           |
| 33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513') |                                             |       |           |
| 15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641') |                                             |       |           |
| 34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507') |                                             |       |           |
| 16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444') |                                             |       |           |
| 8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO        |                                             |       |           |

|                                                 |                                             |
|-------------------------------------------------|---------------------------------------------|
| <b>RKSI(ICN) 23ft</b>                           | <b>VTCC(CNX) 1036ft</b>                     |
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> CNX Traffic 131.5<br>D-ATIS Avail |

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

|              |                                   |                       |                                   |                    |
|--------------|-----------------------------------|-----------------------|-----------------------------------|--------------------|
| 33L/R        | BOPTA xA                          | HD 333 ALT ATC        |                                   |                    |
| 34L/R        | BOPTA xY                          | HD 333 ALT ATC        |                                   |                    |
| 15L/R        | BOPTA xC                          | HD 153 ALT 5000 NADP2 |                                   |                    |
| 16L/R        | BOPTA xH                          | HD 153 ALT 5000 NADP2 |                                   |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9     | 15L ISLL<br>111.9                 | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1     | 16L IRKS<br>110.35                | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                       | 34L/R : EO34L/R, R242<br>YJU R271 |                    |

Parallel TWY 10KTS 이상(R17 MAX 15kts)

[FUK 127.5](#)([SENKA /20](#))  
[TPE 125.5 – 129.1 – HKG 132.15 – 127.1](#)



CNX [\[001/181\]](#) : TL130 UTC+7, **Caution TERR**  
**BKK to CNX APP 80NM, FL160 – Rep PSN, ALT, SQ**  
**(SPD 220kts 20-25 TRK Mile, 180kts IF, 150-160kts FAF)**

|     |              |       |          |
|-----|--------------|-------|----------|
| 36  | MONLO xx A   | PUKAT | ILS Z 36 |
| 18  | MONLO xx B   | PILEX | RNP 18   |
| NAV | CMA, 36 ICMA |       |          |

36 : E(6558', 90도), D(7483')  
18 **DIS** : F(6853', 90도) , G(9199', 90도)  
Narrow RWY, No TWY CL L/T, Yellow Center Line

Difference VDGS, After LD CONF 1

# Meter/Feet Conversion Table

Westbound



Eastbound



**13,100 m**

**43,000 ft**

12,200 m

40,100 ft

**11,600 m**

**38,100 ft**

11,000 m

36,100 ft

**10,400 m**

**34,100 ft**

9,800 m

32,100 ft

**9,200 m**

**30,100 ft**

8,400 m

27,600 ft

**7,800 m**

**25,600 ft**

7,200 m

23,600 ft

**6,600 m**

**21,700 ft**

6,000 m

19,700 ft

**5,400 m**

**17,700 ft**

4,800 m

15,700 ft

**4,200 m**

**13,800 ft**

**3,600 m**

**11,800 ft**

**3,000 m**

**9,800 ft**

2,400 m

7,900 ft

**1,800 m**

**5,900 ft**

1,200 m

3,900 ft

13,700 m

44,900 ft

**12,500 m**

**41,100 ft**

11,900 m

39,100 ft

**11,300 m**

**37,100 ft**

10,700 m

35,100 ft

**10,100 m**

**33,100 ft**

9,500 m

31,100 ft

**8,900 m**

**29,100 ft**

8,100 m

26,600 ft

**7,500 m**

**24,600 ft**

6,900 m

22,600 ft

**6,300 m**

**20,700 ft**

5,700 m

18,700 ft

**5,100 m**

**16,700 ft**

4,500 m

14,800 ft

**3,900 m**

**12,800 ft**

3,300 m

10,800 ft

**2,700 m**

**8,900 ft**

2,100 m

6,900 ft

**1,500 m**

**4,900 ft**

## ALT/HEIGHT Conversion

1,000 m

3,300 ft

500 m

1,600 ft

**900 m**

**3,000 ft**

**450 m**

**1,500 ft**

800 m

2,600 ft

400 m

1,300 ft

**700 m**

**2,300 ft**

**350 m**

**1,100 ft**

600 m

2,000 ft

300 m

1,000 ft

**550 m**

**1,800 ft**

**VTCC(CNX) 1036ft** **RKSI(ICN) 23ft**

CNX Traffic 131.5  
-5min, By Voice GND

**PA**

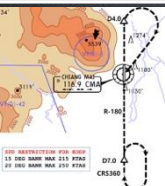
KE ICN 131.5

**CNX : RNAV SID (NADP 1) TA 11000**

**After Airborne L TURN DCT MONLO then R-207**

|           |                      |                                              |
|-----------|----------------------|----------------------------------------------|
| 36        | MONLO xN             | HD 001 ALT 11000 SPD 250<br>TA 11000         |
| 18        | MONLO xS             | HD 181 ALT 6000(Init CL)<br>SPD 230 TA 11000 |
| CMA 116.9 |                      | 36 ICMA 109.9                                |
| FIX       | 36 : CMA /4,/7, R180 | 18 : CMA /7, R180                            |

CHK CLB Gradient, RECMD FLEX Due EGT Limit



[DEP 129.6 – BKK 133.1](#)

[VIE 128.3 – HNI 123.3](#)

[SNY 122.6 – HKG 127.1 – 125.35](#)

[TPE 129.1\(126.7, 127.9\) – 125.5](#)

[FUK 127.5 \(SENKA /20\)](#)

[ICN 125.725\(124.52\) – 120.72](#)



**ICN [333/153] : STAR**

|       |                                                                                            |       |           |
|-------|--------------------------------------------------------------------------------------------|-------|-----------|
| 33/34 | OLMEN xE                                                                                   | ENPIL | OLMEN 180 |
| 15/16 | OLMEN xH                                                                                   | MUNAN | OLMEN 180 |
| NAV   | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR<br>WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |       |           |
| FIX   | RWY /8( 180kts), /5(160kts), YJU R271                                                      |       |           |

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')  
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')  
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts 이상, **HIRO**

|                        |                       |
|------------------------|-----------------------|
| <b>RJAA(NRT) 135ft</b> | <b>PHNL(HNL) 13ft</b> |
|------------------------|-----------------------|

KE Tokyo 131.70  
DCL -15분

**PA**

AOS Oper 132.0  
No D-ATIS

NRT : SID – NADP 1

|              |                  |                     |                  |                  |
|--------------|------------------|---------------------|------------------|------------------|
| 16L/R        | GULBO x          | HD 158 ALT ATC      |                  |                  |
| 34L/R        |                  | HD 338 ALT 7000/ATC |                  |                  |
| NRE<br>117.9 | 16L ITM<br>110.7 | 16R IKF<br>111.5    | 34L IYQ<br>111.9 | 34R ITJ<br>110.9 |

APU Start, TAXI RTE 1, 2, 3, 4 RWY 별 DEP RTE

DEP 124.2

TKO 120.5 – 133.45



HNL [042/079/259] UTC-10 TL180 CPDLC RUJ-KZAK  
Before SYVAD/DANNO/CANON/THOMA CTC HNL

|     |                                            |                |
|-----|--------------------------------------------|----------------|
| 08L | KLANI/BOOKE/OPACA<br>Tx Modify/RDR Vector  | ILS 08L        |
| 04R |                                            | ILS Y 04R      |
| 26L |                                            | RNP AR/LDA 26L |
| NAV | HNL, 08L IHNL, 04R IIUM, (26L IEPC offset) |                |
| FIX | RWY /8                                     |                |

08L : Y(6837'), H(7887'), 26L : R5(7759')

04R : END of RWY Except ATC, NO LAHSO to ATC

AFT RWY 08L L/D: TWY H to A Caution  
RAMP IN West/East Guide Line by ATC



# EDTO Procedure

## PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning  
(Ops Pecs C055) -> EDTO ERA Only(ERA no Wx)

RVSM CHK : CAPT/FO 20ft, PILOT/FE 75ft  
(PFD/ISIS 60ft, ST-BY 300ft)

NAV DATA Input : EEP, ETP1, ETP2, EXP

HF SELCAL CHK : Jeppesen - ENT DATA Pacific

SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

## AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : CAPT/FO 200ft

## BEFORE EEP (Entry Point, ERA 기준)

60min(Threshold) : A330 420NM

One Eng : CRZ SPD 310kts, 180min 1235NM

FIX 1 : EEP, FIX 2 : ETP1

FMS ALT A/P SET : DATA – ETP Page

EDTO C/L : Fuel, A/C, MSA, ALT Wx & NOTAM

## Review Contingency Procedure

- Drift Down 30도이상, 5NM, FL290이하, +-500ft

- Wx Dev 5NM 이상, +-300ft

## EDTO Segment

Apply Actual Wx for Actual Divert

## Diversion

- Enroute RNP downgrade

- Weather min ERA AP

- LAND ASAP by ECAM, QRH

- Only 1 GEN(IDG, APU, EMER) + Multi Fail

- Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)

- Increased Fuel Consumption

## ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET

EDTO C/L : Fuel, A/C, MSA, ALT Wx & NOTAM

Last ETP(Critical Point) Fuel less then PLAN –

Continue by PIC

## EXP (Exit Point)

|                                                                                                                  |                                                                            |                         |                 |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------|-----------------|
| PHNL(HNL) 13ft                                                                                                   |                                                                            | RJAA(NRT) 135ft         |                 |
| AOS Oper 132.0<br>By Voice, No D-ATIS                                                                            |                                                                            | PA                      | KE Tokyo 131.70 |
| HNL : UTC-10, TA 18000, CPDLC KZAK                                                                               |                                                                            |                         |                 |
| 08R/L                                                                                                            | KEOLA x<br>Tx By Plan                                                      | HD 079 ALT 5000 TA18000 |                 |
| 26L/R                                                                                                            |                                                                            | HD 259 ALT 5000 TA18000 |                 |
| HNL 114.8                                                                                                        |                                                                            | 08L IHNL 111.7          | 04R IIUM 110.5  |
| FIX                                                                                                              | 08R : HNL 079/ 1.3, 08L : HNL R045<br>04L/R : HNL /2 then CRS 160          |                         |                 |
| Gate Hold Proc JEPP<br>PushBack & Start to GND, Gate Start to RAMP                                               |                                                                            |                         |                 |
|                                  |                                                                            |                         |                 |
| HCF 118.3                                                                                                        |                                                                            |                         |                 |
|                              |                                                                            |                         |                 |
| NRT [337/157] : LESPO 150                                                                                        |                                                                            |                         |                 |
| 34L/R                                                                                                            | SUPOK E<br>(SUPOK T)                                                       | ELGAR<br>(TYLER)        | ILS 34L/R(Z)    |
| 16L/R                                                                                                            | SUPOK G<br>(SUPOK N)                                                       | GEMIN<br>(NORMA)        | ILS Z 16L/R     |
| NAV                                                                                                              | NRE, 16L ITM, 16R IKF, 34L IYQ, 34R ITJ                                    |                         |                 |
| FIX                                                                                                              | 16L : ITM 4 / 34R : ITJ 14, 4 (DME)<br>16R : IKF 4 / 34L : IYQ 12, 4 (DME) |                         |                 |
| 16L : B6(6433'), B7(7017'), 34R : B4(5849'), B2(6778')<br>16R : A6(6076'), A7(7624'), 34L : A5(6167'), A4(7641') |                                                                            |                         |                 |
| L/D DOWN before 14/12 DME, L/D FLAP 4 DME<br>Arrival Taxi RTE in Jeppesen (No Numbering)                         |                                                                            |                         |                 |

|                       |                       |
|-----------------------|-----------------------|
| <b>RKSI(ICN) 23ft</b> | <b>YSSY(SYD) 21ft</b> |
|-----------------------|-----------------------|

|                                                  |                                          |
|--------------------------------------------------|------------------------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT 5분 차이시<br>CTC Comm | <b>PA</b> Menzies SYD 129.6<br>No D-ATIS |
|--------------------------------------------------|------------------------------------------|

|                                        |
|----------------------------------------|
| ICN : SID (33/34 NADP 1, 15/16 NADP 2) |
|----------------------------------------|

|              |                                   |                               |                                  |                    |
|--------------|-----------------------------------|-------------------------------|----------------------------------|--------------------|
| 33L/R        | OSPOT xE/A                        | HD 333 ALT 5500/ATC SPD 230/x |                                  |                    |
| 34L/R        | OSPOT xY                          | HD 333 ALT ATC SPD 230        |                                  |                    |
| 15L/R        | OSPOT xC                          | HD 153 ALT 5000 NADP2         |                                  |                    |
| 16L/R        | OSPOT xH                          | HD 153 ALT 5000 NADP2         |                                  |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9             | 15L ISLL<br>111.9                | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1             | 16L IRKS<br>110.35               | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                               | 34L/R : EO34L/R,R242<br>YJU R271 |                    |

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

|                                                                                                                |                                                                                      |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <a href="#">TGU 125.37</a><br><a href="#">Kobe 118.9 – FUK APP 119.65</a><br><a href="#">FUK RDR – 121.125</a> |  |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

|                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------|
| SYD [335/155] UTC+10/11 TL110 RW34L/16R on Req<br>CHK SEMAP, Curfew 2300-0600LT, CPDLC PSN rep<br>ILS PRM, Noise Abate Proc Review |
|------------------------------------------------------------------------------------------------------------------------------------|

|       |                                        |       |           |
|-------|----------------------------------------|-------|-----------|
| 16L/R | BOREE x A                              | URDEN | ILS 16L/R |
| 34R/L | (x P for 16R)                          | SOSIJ | ILS 34L/R |
| NAV   | SY, 34L ISN, 34R IKN, 16L ISS, 16R IKS |       |           |
| FIX   | RW /10(185-160kt), /5(160-150kts)      |       |           |

|                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 34R : U1(5515'), L(7378'), 34L: A3(6207'), A2(6620')<br>16LDis:T4(5718'),T6(6971'), 16R : A3(6210'),A4(6561')<br>Prefer Exit Unble ATC, min REVTH, NO LASHO |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                   |
|-----------------------------------------------------------------------------------|
| SY 20NM Below 9000ft, Intl x, Dom x - TWY<br>Same Spot no. International/Domestic |
|-----------------------------------------------------------------------------------|

# EDTO Procedure

## PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning  
(Ops Pecs C055) -> EDTO ERA Only(ERA no Wx)

RVSM CHK : CAPT/FO 20ft, PILOT/FE 75ft  
(PFD/ISIS 60ft, ST-BY 300ft)

NAV DATA Input : EEP, ETP1, ETP2, EXP

HF SELCAL CHK : Jeppesen - ENT DATA Pacific

SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

## AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : CAPT/FO 200ft

## BEFORE EEP (Entry Point, ERA 기준)

60min(Threshold) : A330 420NM

One Eng : CRZ SPD 310kts, 180min 1235NM

FIX 1 : EEP, FIX 2 : ETP1

FMS ALT A/P SET : DATA – ETP Page

EDTO C/L : Fuel, A/C, MSA, ALT Wx & NOTAM

## Review Contingency Procedure

- Drift Down 30도이상, 5NM, FL290이하, +-500ft

- Wx Dev 5NM 이상, +-300ft

## EDTO Segment

Apply Actual Wx for Actual Divert

TERR Escape RTE Papua New Guinea

## Diversion

- Enroute RNP downgrade

- Weather min ERA AP

- LAND ASAP by ECAM, QRH

- Only 1 GEN(IDG, APU, EMER) + Multi Fail

- Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)

- Increased Fuel Consumption

## ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET

EDTO C/L : Fuel, A/C, MSA, ALT Wx & NOTAM

Last ETP(Critical Point) Fuel less then PLAN –  
Continue by PIC

## EXP (Exit Point)

YSSY(SYD) 21ft

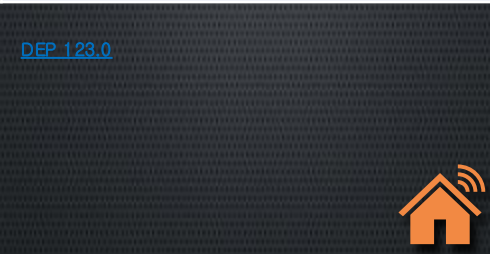
RKSI(ICN) 23ft

|                                                             |                           |
|-------------------------------------------------------------|---------------------------|
| Menzies SYD 129.6<br>No ATIS, Voice by<br>CLR133.8/GND121.7 | <b>PA</b><br>KE ICN 131.5 |
|-------------------------------------------------------------|---------------------------|

SYD : UTC+10/11, TA 10000,  
CPDLC YMMM on GND 45min/Passing 10M'

|          |                               |                  |                                    |                  |
|----------|-------------------------------|------------------|------------------------------------|------------------|
| 16R      | GROOK x RIC tx<br>(SYD x RDR) |                  | HD 155 ALT ATC SPD 270<br>TA 10000 |                  |
| 34L      | RIC x<br>(SYD x RDR)          |                  | HD 335 ALT ATC<br>TA 10000         |                  |
| SY 112.1 | 34L ISN<br>110.1              | 16R IKS<br>109.5 | 34R IKN<br>109.3                   | 16L ISS<br>110.9 |

CASA no approval KAL (for T/O min)  
CHK Special Requirement SID Chart



|                                                        |                                             |       |           |
|--------------------------------------------------------|---------------------------------------------|-------|-----------|
| ICN [333/153] : STAR                                   |                                             |       |           |
| 33/34                                                  | GUKDO xE                                    | ENPIL | GUKDO 180 |
| 15/16                                                  | GUKDO xH                                    | MUNAN | GUKDO 180 |
| NAV                                                    | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR |       |           |
|                                                        | WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |       |           |
| FIX                                                    | RWY /8(180kts), /5(160kts) , YJU R271       |       |           |
| 33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513') |                                             |       |           |
| 15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641') |                                             |       |           |
| 34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507') |                                             |       |           |
| 16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444') |                                             |       |           |
| 8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO        |                                             |       |           |

|                       |                       |
|-----------------------|-----------------------|
| <b>RKSI(ICN) 23ft</b> | <b>YBBN(BNE) 15ft</b> |
|-----------------------|-----------------------|

|                                                  |                              |
|--------------------------------------------------|------------------------------|
| KE ICN 131.5<br>DCL -10분 TOBT 5분 차이시<br>CTC Comm | <b>PA</b> Menzies BNE 132.55 |
|--------------------------------------------------|------------------------------|

ICN : SID (33/34 NADP 1, 15/16 NADP 2)

|              |                                   |                               |                                  |                    |
|--------------|-----------------------------------|-------------------------------|----------------------------------|--------------------|
| 33L/R        | OSPOT xE/A                        | HD 333 ALT 5500/ATC SPD 230/x |                                  |                    |
| 34L/R        | OSPOT xY                          | HD 333 ALT ATC SPD 230        |                                  |                    |
| 15L/R        | OSPOT xC                          | HD 153 ALT 5000 NADP2         |                                  |                    |
| 16L/R        | OSPOT xH                          | HD 153 ALT 5000 NADP2         |                                  |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9             | 15L ISLL<br>111.9                | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1             | 16L IRKS<br>110.35               | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                               | 34L/R : EO34L/R,R242<br>YJU R271 |                    |

Parallel TWY 10KTS 이상(R17 MAX 15kts)

[TGU 125.37](#)  
[Kobe 118.9 – FUK APP 119.65](#)  
[FUK RDR – 121.125](#)



BNE [016/196] UTC+10 TL110 Ref ALT A/P VBCG LD WT  
 REF SPD REST in CHART, CHK PRM Procedure

|       |                                        |             |           |
|-------|----------------------------------------|-------------|-----------|
| 01L/R | SMOKA x A                              | TANIK/GLENN | ILS 01L/R |
| 19R/L |                                        | OSOKO/BETSO | ILS 19R/L |
| NAV   | BN, 01L IBN, 19R IBE, 01R IBA, 19L IBS |             |           |
| FIX   | RW xx : /5(160-150kts) into FIX Avail  |             |           |

01L : T6(6243'), T4(7598'), 19R : T9(6246'), T11(7595')  
 01R : A4(7043'), A3(8579'), 19L : A6(5620'), A7(8477')  
 Prefer Exit Unble ATC, min REV TH, T5, T10 No Exit

TWY B for L/D, TAXI with IDLE(Min PWR)  
 Short Ramp-In Lead Line, Slow Down before Turning

# EDTO Procedure

## PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning  
(Ops Pecs C055) -> EDTO ERA Only(ERA no Wx)

RVSM CHK : CAPT/FO 20ft, PILOT/FE 75ft  
(PFD/ISIS 60ft, ST-BY 300ft)

NAV DATA Input : EEP, ETP1, ETP2, EXP

HF SELCAL CHK : Jeppesen - ENT DATA Pacific

SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

## AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : CAPT/FO 200ft

## BEFORE EEP (Entry Point, ERA 기준)

60min(Threshold) : A330 420NM

One Eng : CRZ SPD 310kts, 180min 1235NM

FIX 1 : EEP, FIX 2 : ETP1

FMS ALT A/P SET : DATA – ETP Page(Nearest AP)

EDTO C/L : Fuel, A/C, MSA, ALT Wx & NOTAM

## Review Contingency Procedure

- Drift Down 30도이상, 5NM, FL290이하, +-500ft
- Wx Dev 5NM 이상, +-300ft

## EDTO Segment

Apply Actual Wx for Actual Divert

TERR Escape RTE Papua New Guinea

## COMM (CPDLC)

RJJJ – KZAK(FIR) – AYPM(FIR) – YBBB(FIR)

(Cross GUM airspace 250NM 118.7, SQ2100)

## Diversion

- Enroute RNP downgrade
- Weather min ERA AP
- LAND ASAP by ECAM, QRH
- Only 1 GEN(IDG, APU, EMER) + Multi Fail
- Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)
- Increased Fuel Consumption

## ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET

EDTO C/L : Fuel, A/C, MSA, ALT Wx & NOTAM

Last ETP(Critical Point) Fuel less then PLAN –  
Continue by PIC

## EXP (Exit Point)

|                                      |                       |
|--------------------------------------|-----------------------|
| <b>YBBN(BNE)15ft</b>                 | <b>RKSI(ICN) 23ft</b> |
| <b>PA</b>                            |                       |
| Menzies BNE 132.55<br>Voice by -5min | KE ICN 131.5          |

BNE : UTC+10, TA 10000,  
CPDLC YBBB on GND 45min/Passing 10M'

|          |                                                          |                                            |                  |                  |
|----------|----------------------------------------------------------|--------------------------------------------|------------------|------------------|
| 01L/R    | BIXAD x<br>(01R Diff Chart)<br>BNE x<br>RDR SID with H/D | HD 016 ALT ATC(6000)<br>SPD x/240 TA 10000 |                  |                  |
| 19R/L    |                                                          | HD 196 ALT ATC(6000)<br>SPD 240 TA 10000   |                  |                  |
| BN 113.2 | 01L IBN<br>111.5                                         | 19R IBE<br>109.3                           | 01R IBA<br>109.5 | 19L IBS<br>110.1 |

PushBack Point C,D,E , TWY A for T/O  
Do Not PASS B1 for 19L  
“CNL SID, TRK EXT CL, xx Degree” -> Main RWY TRK

DEP 133.45 –



| ICN [333/153] : STAR                                   |                                             |       |           |
|--------------------------------------------------------|---------------------------------------------|-------|-----------|
| 33/34                                                  | GUKDO xE                                    | ENPIL | GUKDO 180 |
| 15/16                                                  | GUKDO xH                                    | MUNAN | GUKDO 180 |
| NAV                                                    | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR |       |           |
|                                                        | WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS |       |           |
| FIX                                                    | RWY /8(180kts), /5(160kts) , YJU R271       |       |           |
| 33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513') |                                             |       |           |
| 15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641') |                                             |       |           |
| 34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507') |                                             |       |           |
| 16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444') |                                             |       |           |
| 8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO        |                                             |       |           |



|                       |                         |
|-----------------------|-------------------------|
| <b>RKSI(ICN) 23ft</b> | <b>ZMCK(UBN) 4485ft</b> |
|-----------------------|-------------------------|

|                                                 |           |                   |
|-------------------------------------------------|-----------|-------------------|
| KE ICN 131.5<br>DCL -10분 TOBT5분 차이시<br>CTC Comm | <b>PA</b> | None<br>No D-ATIS |
|-------------------------------------------------|-----------|-------------------|

|                                        |
|----------------------------------------|
| ICN : SID (33/34 NADP 1, 15/16 NADP 2) |
|----------------------------------------|

|              |                                   |                       |                                  |                    |
|--------------|-----------------------------------|-----------------------|----------------------------------|--------------------|
| 33L/R        | NOPIK xA                          | HD 333 ALT ATC        |                                  |                    |
| 34L/R        | NOPIK xY                          | HD 333 ALT ATC        |                                  |                    |
| 15L/R        | BINIL xC                          | HD 153 ALT 5000 NADP2 |                                  |                    |
| 16L/R        | BINIL xH                          | HD 153 ALT 5000 NADP2 |                                  |                    |
| NCN<br>113.8 | 33L INLL<br>109.3                 | 33R INRR<br>108.9     | 15L ISLL<br>111.9                | 15R ISRR<br>109.1  |
| WNG<br>112.9 | 34L IRFN<br>109.95                | 34R IRKN<br>108.1     | 16L IRKS<br>110.35               | 16R IRFS<br>108.55 |
| FIX          | 33L/R : NC05L/R, R242<br>YJU R271 |                       | 34L/R : EO34L/R,R242<br>YJU R271 |                    |

|                                      |
|--------------------------------------|
| Parallel TWY 10KTS 이상(R17 MAX 15kts) |
|--------------------------------------|

|                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>DEP 125.15 – TGU 132.8 – DLC 132.95</div> <div>  </div> |
|--------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| UBN [110/290] TL 148 : Hi FE,TERR/Restrict Area<br>CTC UBT CTL 5min Before FIR(POLHO, INTIK)<br>UBT CTL ask L/D Time, Hi FE & Sink Rate due TAS |
|-------------------------------------------------------------------------------------------------------------------------------------------------|

|     |             |       |              |
|-----|-------------|-------|--------------|
| 29  | MENUG xx W  | ZONAM | ILS X 29     |
| 11  | MENUG xx V  | CK113 | RNP 11 (Req) |
| NAV | SER, 29 CDA |       |              |

|                                                  |
|--------------------------------------------------|
| 29 : C(5761’), B(8241’), 11 : C(5761’), D(8231’) |
|--------------------------------------------------|

|                                     |
|-------------------------------------|
| Gate 2 A330 Must Follow Dashed Line |
|-------------------------------------|

| Meter/Feet Conversion Table                                                                    |                  |                                                                                                |                  |
|------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------|------------------|
| Westbound<br> |                  | Eastbound<br> |                  |
|                                                                                                |                  | 13,700 m                                                                                       | 44,900 ft        |
| <b>13,100 m</b>                                                                                | <b>43,000 ft</b> | <b>12,500 m</b>                                                                                | <b>41,100 ft</b> |
| 12,200 m                                                                                       | 40,100 ft        | 11,900 m                                                                                       | 39,100 ft        |
| <b>11,600 m</b>                                                                                | <b>38,100 ft</b> | <b>11,300 m</b>                                                                                | <b>37,100 ft</b> |
| 11,000 m                                                                                       | 36,100 ft        | 10,700 m                                                                                       | 35,100 ft        |
| <b>10,400 m</b>                                                                                | <b>34,100 ft</b> | <b>10,100 m</b>                                                                                | <b>33,100 ft</b> |
| 9,800 m                                                                                        | 32,100 ft        | 9,500 m                                                                                        | 31,100 ft        |
| <b>9,200 m</b>                                                                                 | <b>30,100 ft</b> | <b>8,900 m</b>                                                                                 | <b>29,100 ft</b> |
| 8,400 m                                                                                        | 27,600 ft        | 8,100 m                                                                                        | 26,600 ft        |
| <b>7,800 m</b>                                                                                 | <b>25,600 ft</b> | <b>7,500 m</b>                                                                                 | <b>24,600 ft</b> |
| 7,200 m                                                                                        | 23,600 ft        | 6,900 m                                                                                        | 22,600 ft        |
| <b>6,600 m</b>                                                                                 | <b>21,700 ft</b> | <b>6,300 m</b>                                                                                 | <b>20,700 ft</b> |
| 6,000 m                                                                                        | 19,700 ft        | 5,700 m                                                                                        | 18,700 ft        |
| <b>5,400 m</b>                                                                                 | <b>17,700 ft</b> | <b>5,100 m</b>                                                                                 | <b>16,700 ft</b> |
| 4,800 m                                                                                        | 15,700 ft        | 4,500 m                                                                                        | 14,800 ft        |
| <b>4,200 m</b>                                                                                 | <b>13,800 ft</b> | <b>3,900 m</b>                                                                                 | <b>12,800 ft</b> |
| 3,600 m                                                                                        | 11,800 ft        | 3,300 m                                                                                        | 10,800 ft        |
| <b>3,000 m</b>                                                                                 | <b>9,800 ft</b>  | <b>2,700 m</b>                                                                                 | <b>8,900 ft</b>  |
| 2,400 m                                                                                        | 7,900 ft         | 2,100 m                                                                                        | 6,900 ft         |
| <b>1,800 m</b>                                                                                 | <b>5,900 ft</b>  | <b>1,500 m</b>                                                                                 | <b>4,900 ft</b>  |
| 1,200 m                                                                                        | 3,900 ft         |                                                                                                |                  |

| ALT/HEIGHT Conversion |                 |              |                 |
|-----------------------|-----------------|--------------|-----------------|
| 1,000 m               | 3,300 ft        | 500 m        | 1,600 ft        |
| <b>900 m</b>          | <b>3,000 ft</b> | <b>450 m</b> | <b>1,500 ft</b> |
| 800 m                 | 2,600 ft        | 400 m        | 1,300 ft        |
| <b>700 m</b>          | <b>2,300 ft</b> | <b>350 m</b> | <b>1,100 ft</b> |
| 600 m                 | 2,000 ft        | 300 m        | 1,000 ft        |
| <b>550 m</b>          | <b>1,800 ft</b> |              |                 |

ZMCK(UBN) 4485ft

RKSI(ICN) 23ft

None

No D-ATIS

CLR by Voice to TWR 129.0

PA

KE ICN 131.5

UBN : SID (NADP 1) TA 128

|           |                                                                                                 |                         |
|-----------|-------------------------------------------------------------------------------------------------|-------------------------|
| 29        | MENUG x C/A                                                                                     | HD 290 ALT ATC TA 12800 |
| 11        | MENUG x D/B                                                                                     | HD 110 ALT ATC TA 12800 |
| SER 116.9 |                                                                                                 | 29 CDA109.9             |
| FIX       | 29 : SER 290/4, R343, SER R313, SER 290/10, R260<br>11: SER 110/1.5, SER R313, SER 290/10, R260 |                         |

TWY F very Narrow, TWY B,D Opposite Turn Prohibit



DEP 120.0

ICN 132.8 – APP 119.75



ICN [333/153] : STAR

|       |                                              |       |           |
|-------|----------------------------------------------|-------|-----------|
| 33/34 | REBIT x A                                    | PAMBI | REBIT 170 |
| 15/16 | REBIT x H                                    | MUNAN | REBIT 170 |
| NAV   | NCN, 33L INLL, 33R INRR, 15L ISLL, 15R ISRR  |       |           |
|       | WNG, 34L IRFN, 34R IRKN, 16L IRKS, 16R IRFS  |       |           |
| FIX   | RWY /8(180kts), /5(160kts) , P518 R068, R278 |       |           |

33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')

15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')

16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

# Differences (ATS A333)

|                                                       |              |                                 |                                            |
|-------------------------------------------------------|--------------|---------------------------------|--------------------------------------------|
| <b>A322(5)</b><br>HL7524-7551                         | APMS         | EY 252 (151)<br>EY 260 (156)    | Total<br>PAX 200 NOR<br><br>CHK OFP PAX No |
| <b>A323(11)</b><br>HL7553-7720                        |              |                                 |                                            |
| <b>A223E(3)</b><br>HL82xx                             | APMS<br>-2.5 | EY 188 (112)                    | HL 80xx<br>PAX AUTO                        |
| <b>A323E(6)</b><br>HL80xx<br>RNP AR, OANS<br>PAX AUTO | APMS<br>-1.5 | EY 248 (148)<br>HL8001 RNP AR X | Enhanced<br>TRIM AUTO                      |
| ENG START                                             |              | HL7xxx 43-48%                   | HL8xxx 52-54%                              |

## PRINCIPAL DIMENSION

WINGSPAN : ICAO E, FAA V  
Weight CAT: Heavy

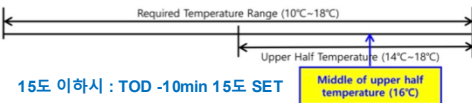
|             | A330-300 / 300E |           | A330-200 / 200E      |                        |
|-------------|-----------------|-----------|----------------------|------------------------|
| LENGTH      | 63.69 m         | 208.92 ft | 58.37 m              | 191.25 ft              |
| WING SPAN   | 60.30 m         | 197.83 ft | 60.30 m              | 197.83 ft              |
| HEIGHT      | 16.83 m         | 55.25 ft  | 17.80 m /<br>17.30 m | 58.42 ft /<br>56.67 ft |
| WHEEL BASE  | 25.37 m         | 83.25 ft  | 22.18 m              | 72.75 ft               |
| WHEEL TREAD | 10.68 m         | 35.08 ft  | 10.68 m              | 35.08 ft               |

Refer to FCOM DSC-20-20 Principal Dimensions · 180 Back 72도 48m/(157ft)

## MTOW CAT ST-IN : C, Circling C/D(FAA)

|      |             |      |             |
|------|-------------|------|-------------|
| 238T | 524,700 lbs | 230T | 507,063 lbs |
| 217T | 478,402 lbs | 198T | 436,515 lbs |
| 192T | 423,287 lbs |      |             |

## LOADSHEET : CARGO No. FWD 11-21, AFT 31-41



## CRZ FUEL Penalty (Approximation)

### A330

4000ft below OPT ALT : 5% increase trip fuel  
8000ft below OPT ALT : 10% increase trip fuel  
Above OPT ALT : 3% increase trip fuel

## FUEL Consumption

### APU

GND : **438 LBS/HR** (BLOCK FUEL산정)  
IN FLT : **286 LBS/HR**

### TAXI

2 ENG, no APU : 3300 LBS/HR (**10분 550 LBS**)

### CRZ

- 100FT 당 8 MIN  
ex) **1000FT 상승 80 MIN, 1HR 20MIN**

### Holding

분당 **200LBS** (Actual F/F 참조, 시간당 **12000LBS**)

### Missed App & Landing

**3000 LBS** (과거 EDTO자료 1회 Missed App+L/D)

## FUEL Loading

### Tail Tank :

CL TANK 80000LBS 이상 5000LBS 까지  
CL TANK 160000LBS 이상 10000 LBS 까지

FUEL Overfill : 1000LBS 기준



## PACKS USE WITH LP AIR CONDITION UNIT (SD 연결 표시없음)

공항 요구로 APU OFF후 기내 온도 조절을 위한 방법  
Low Press Unit을 연결하며 APU와 동시사용 금지

**The flight crew must not use conditioned air  
from the packs and from the LP Air**

**Conditioning Unit at the same time, to prevent  
any adverse effect on the Air Conditioning  
system.**

절차 POM, FCOM에 없음

APU BLEED OFF 후 LP AIR 연결

EXT PWR 연결후 APU OFF

APU Start - LP Disconnect - APU BLEED ON

## APU BLEED USE WITH HP AIR START UNIT (SD 연결 표시없음)

APU 부작동시 AIR START UNIT으로 시동을 위해 사용  
외부 BLEED AIR의 역할을 함. APU BLEED 동시사용  
금지

**The flight crew must not use bleed air from  
the APU BLEED and from the HP Air Start Unit  
at the same time, to prevent any adverse  
effect on the Bleed Air System.**

● Before connecting the air start unit:

**PACK 1 & 2 ..... as REQ F/O**

HP AIR 시동전 Air Condition으로 사용시(시동시 OFF)

**APU BLEED ..... OFF F/O**

**ENG 1 & 2 BLEED ..... OFF F/O**

**X BLEED ..... OPEN F/O**

**AIR START UNIT CONNECT ... REQ CAP**

Two ground air start units may be used in parallel if the  
pressure/flow relation is expected to be marginal.



**ENG START  
Next Page**

# STARTING with AIR START UNIT

통상 ENG 1 먼저 (EXT PWR Start 2 ENG)

“Req Engine Start up Present Positon~~~”

## ● Before connecting the air start unit:

**PACK 1 & 2 ..... OFF F/O**

prevent any possible contamination of the packs by the air start unit.

**APU BLEED ..... OFF F/O**

**ENG 1 & 2 BLEED ..... OFF F/O**

**X BLEED ..... OPEN F/O**

**AIR START UNIT CONNECT ... REQ CAP**

Two ground air start units may be used in parallel if the pressure/flow relation is expected to be marginal.

## ● When cleared to start:

**Note:** The **AIR ABNORM BLEED CONFIG** alert is triggered after the first engine start. It can be disregarded.

**ENG 1 ..... START CAP**

**Note:** As necessary, engine 2 can be started first.

In this case, check the brake ACCU pressure prior to engine start.

**Note:** The minimum recommended starter air supply pressure is 25 PSI when the start valve is open.

## ● After engine 1 is started:



- If the air start unit is used to start engine 2:

**ENG 2 ..... START CAP**

- When engine 2 is started:

**AIR START UNIT DISC . . . . REQ CAP**

**X BLEED ..... AUTO F/O**

**ENG 1 & 2 BLEED ..... ON F/O**

**PACK 1 & 2 ..... ON F/O**

- If the crossbleed engine start procedure is used to start engine 2:

**AIR START UNIT (EXT PWR) DISC. RQ CAP**

**ENG 1 BLEED ..... ON F/O**

**PACK 1 & 2 ..... ON F/O**

**CRSBLD ENG START PROC. APPLY**

**EXT PWR ..... CHK AVAIL CAP**

**EXT PWR DISCONNECT ..... REQ CAP**

**NOTE :** EXT PWR removed after 1 ENG start.

**but FCOM :** In order to ensure that the entire network remains electrically supplied, it is recommended to start both engines before the disconnection of the external electrical power.

**NOR PROC - AFTER START ..... RESUME**

# ENG CROSSBLEED START

#1 ENG BLEED 로 #2 ENG START

**Req PushBack**

**PushBack 완료(Parking Brake set)**

One engine must be running in order to supply air for other engine start.

● Before second engine start :

**APU BLEED ..... OFF F/O**

The BLEED valve of the supplying engine reopens and the cross bleed valve closes.

**ENG BLEED (supply engine) ..... ON F/O**

**ENG BLEED (receive engine) ..... OFF F/O**

The bleed valves of receiving engines are closed to avoid reverse flow leakage.

**X BLEED ..... OPEN F/O**

● When cleared to start :

**AREA CLEAR OF OBS ..... CONF CAP**

Ensure increased power jet wake does not constitute any hazard to people or installations behind the aircraft.

**THR LVR (sup eng) . ADJ BLD PRES CAP**

Adjust thrust of supplying engine to obtain an engine bleed pressure of **30 PSI**. If the thrust required to obtain the appropriate engine bleed pressure exceeds **40 % N1**, pay particular attention to the surrounding area.

**RECEIVING ENGINE ..... START ALL**

Apply the normal engine start procedure.

● After start :

**THR LVR (supply engine) ..... IDLE CAP**

**X BLEED ..... AUTO F/O**

**ENG BLEED (receive engine) ..... ON F/O**

**PACK 1 & 2 ..... ON F/O**





## COLD TEMP CORRECTION General

5도 간격은 보수적으로 보간법 적용 됨

**Min** 제외한 모든 고도 수정은 ATC 인가 필요

**Mandatory, Missed App 고도** ATC 사전 인가 없이 금지

반드시 고도 - FE 후의 고도를 보정해야함.

Ex) FE 200ft 공항 : 5000ft는 4800ft만 보정해야함.

### Height Above FE (Feet) 200-800ft

| TEMP | 200 | 300 | 400 | 500 | 600 | 700 | 800 |
|------|-----|-----|-----|-----|-----|-----|-----|
| 0    | 20  | 20  | 30  | 30  | 40  | 40  | 50  |
| -5   | 20  | 30  | 40  | 40  | 50  | 60  | 70  |
| -10  | 20  | 30  | 40  | 50  | 60  | 70  | 80  |
| -15  | 30  | 40  | 50  | 60  | 80  | 90  | 100 |
| -20  | 30  | 50  | 60  | 70  | 90  | 100 | 120 |

### Height Above FE (Feet) 900-5000ft

| TEMP | 900 | 1000 | 1500 | 2000 | 3000 | 4000 | 5000 |
|------|-----|------|------|------|------|------|------|
| 0    | 50  | 60   | 90   | 120  | 170  | 230  | 280  |
| -5   | 70  | 80   | 120  | 160  | 230  | 310  | 390  |
| -10  | 90  | 100  | 150  | 200  | 290  | 390  | 490  |
| -15  | 110 | 120  | 180  | 240  | 360  | 480  | 600  |
| -20  | 130 | 140  | 210  | 280  | 420  | 570  | 710  |

GMP, CJU, PUS next page



## COLD TEMP CORRECTION (Domestic)

Min 은 반드시 수정 (중간 고도 CORRECTION은 PIC 결정)  
Missed App 고도는 ATC 협조 필요

### GMP 32L (261') / 32R (262') / 14R (254')

| 32L/R | 8000 | 5500 | 5300 | 4000 | 2800 | 2300 | 2000 |
|-------|------|------|------|------|------|------|------|
| 0     | 8450 | 5810 | 5600 | 4230 | 2970 | 2440 | 2120 |
| -5    | 8620 | 5930 | 5710 | 4310 | 3030 | 2490 | 2160 |
| -10   | 8780 | 6040 | 5820 | 4390 | 3080 | 2530 | 2200 |
| R14   | 4000 | 2800 | 1400 |      | 4000 |      |      |
| 0     | 4230 | 2970 | 1490 |      | 4230 |      |      |
| -5    | 4310 | 3030 | 1520 |      | 4310 |      |      |
| -10   | 4390 | 3080 | 1540 |      | 4390 |      |      |

### CJU 07 (307') / 25 (296')

|     |      |      |      |    |      |    |      |
|-----|------|------|------|----|------|----|------|
|     | 4000 | 2900 | 1800 | 07 | 8000 | 25 | 6000 |
| 0   | 4220 | 3070 | 1900 |    | 8450 |    | 6340 |
| -5  | 4300 | 3130 | 1940 |    | 8620 |    | 6460 |
| -10 | 4380 | 3180 | 1970 |    | 8780 |    | 6590 |

### PUS 36L(233'),36R(228') / 18L/R (see below)

|       |      |      |      |      |      |      |      |
|-------|------|------|------|------|------|------|------|
| 36L/R | 6000 | 5000 | 3300 | 2100 |      | 6000 |      |
| 0     | 6340 | 5290 | 3490 | 2210 |      | 6340 |      |
| -5    | 6460 | 5390 | 3560 | 2250 |      | 6460 |      |
| -10   | 6580 | 5490 | 3620 | 2290 |      | 6580 |      |
| 18L/R | 6000 | 5000 | 4000 | 2600 | 1700 |      | 6000 |
| 0     | 6340 | 5290 | 4230 | 2760 | 1800 |      | 6340 |
| -5    | 6460 | 5390 | 4310 | 2810 | 1830 |      | 6460 |
| -10   | 6580 | 5490 | 4390 | 2860 | 1870 |      | 6580 |



## COLD Wx Operation 1/2

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :

- visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
- ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

**PRELIMINARY (OAT 3°C 이하)**

**PROBE/WIN HEAT – ON (ENG START후 AUTO)**

**LOGBOOK ----- CHECK**

**MAX TIME TO NEXT ENG ACC ---- DETERMINE**

**TAXI IN TIME – NEXT ENG ACC TIME 15분 이내인지 확인**

**TAXI-OUT(OAT 3°C 이하)**

**TAXI TIME ----- MONITOR**

15분 초과 또는 Eng Vibrations시 절차 수행

**SOP SURFACE COND & AREA ----- CHECK**

**ATC ----- NOTIFY**

**PARKING BRAKE ----- ON**

the aircraft starts to move, immediately retard the thrust levers to IDLE.

**THR LEVERS ----- 50% N1 (NO HOLD TIME)**

**THR LEVERS ----- RETARD TO IDLE**

Repeat this procedure at intervals not longer than 15 min, or if the engine vibrations increase.

**SOP TAXI ----- RESUME**

**BEFORE TAKEOFF**

**FLAPS lever ----- SET FOR TAKEOFF**

**FLAPS ----- CHECK PSN**

**T/O CONFIG pb ----- TEST**

**T/O MEMO ----- CHECK NO BLUE**

**AFTER START checklist ----- COMPLETE**

**BEFORE TAKEOFF checklist -- ACCOMPLISH**

**TAKEOFF(OAT 3°C 이하)**

A final engine acceleration must be performed before takeoff to entirely clean the engine from ice accretion.

**ATC ----- NOTIFY**

**PARKING BRK(OR BRK PEDALS) ----- APPLY**

**THR LEVERS ----- 50% N1 (NO HOLD TIME)**

**THR LEVERS ----- RETARD TO IDLE**

**SOP TAKEOFF ----- RESUME**

**AFTER L/D**

**DO NOT RETRACT FLAPS**

Ice Shedding 절차 반복



## COLD Wx Operation 2/2

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :

- visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
- ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

### PARKING

After engine shutdown and before shutting down electrical supply:

**FLAPS / SLATS ..... CONF FREE OF CONTAM**

Perform a visual inspection to determine if the flaps/slats mechanism is free of contamination.

If necessary, perform decontamination.

**Y ELEC PUMP pb ..... AUTO F/O**

**G ELEC PUMP pb ..... AUTO F/O**

**Y ELEC PUMP pb (guarded) ..... ON F/O**

**G ELEC PUMP pb (guarded) ..... ON F/O**

**SLATS / FLAPS ..... RETRACT F/O**

Monitor slats/flaps retraction on ECAM upper display

When slats and flaps are retracted:

**Y ELEC PUMP pb (guarded) ..... AUTO F/O**

**G ELEC PUMP pb (guarded) ..... AUTO F/O**

**NORMAL PROCEDURE ..... RESUME ALL**



# ENG ON Deicing in ICN

TOBT- 40min CTC KE ICN (사전신청, 결과확인)

ICN Deicing "Deicing Required ENG On Deicing"

ICN Apron "Req Pushback Deicing Zone xx" **SQ2000**

Pad Control Arrange Deicing Pad No. **REQ FULL BODY**

Ice Man Manage Deicing Process

**Gate 시동후 (FLAP UP, FLT CTL CHK 없이)**

**PARKING BRAKE ----- SET**

Report Parking Brake SET - > Ice Man

**BROADBAND XMT sw ----- OFF**

**COMMUNICATION with GND ----- ESTAB**

**DE/ANTI-ICING FLUID TYPE ----- CHECK**

**CABIN PRESS MODE SEL ---- CHK AUTO**

**ENG 1 & 2 BLEED ----- OFF**

Note : **AIR ENG 1+2 BLEED FAULT** – Disregard

**APU BLEED ----- OFF**

**DITCHING pb ----- ON**

Note : **CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV NOT OPEN, COND LAV + GAL VENT FAULT** –

Disregard

**THRUST LEVERS ----- CHK IDLE**

**"A/C PREPARED FOR SPRAYING"**

Report Ready for Deicing - > Ice Man

**START SPRAY REQ DCL(CTC DEL)**

**UPON COMPLETE SPRAY(TIME CHK 1분)**

용액과 마지막 용액 뿌린 시간 받고 적는다.

Holdover Time 결정!!!

**PITOT STATICS (GND Crew) ----- CHK**

**GND EQUIPMENT ----- REMOVE**

**DITCHING pb ----- OFF**

**OUTFLOW V/V - CHK OPEN(ECAM press)**

**TIME CHECK 1분후 Cold Wx**

**ENG 1 & 2 BLEED ----- ON**

**BROADBAND XMT sw ----- ON**

**5분후**

**APU BLEED ----- AS RQRD**

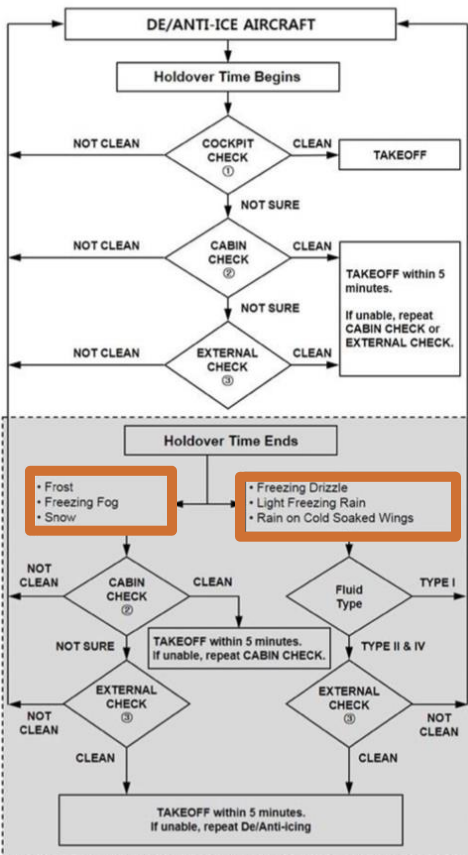
**NORMAL PROCEDURE ----- RESUME**

**FLT CONTROL CHK, FLAPS UP**

**DECISION TREE next page**



# TAKEOFF DECISION TREE



# ENG OFF Deicing in GMP...

## TOBT- 20min CTC KE GMP (PAD, New TOBT)

### REQ DCL

Deicing "Deicing Required PADxxx"  $\pm 5$ min TOBT  
Apron "Req Pushback Deicing PADxxx"

Gate 시동 후 (FLAP UP, FLT CTL CHK, APU OFF 없이)

Engine anti-ice sw ----- OFF

PARKING BRAKE ----- SET

BROADBAND XMT sw ----- OFF

COMMUNICATION with GND ----- ESTAB

DE/ANTI-ICING FLUID TYPE ----- CHECK

CABIN PRESS MODE SEL ---- CHK AUTO

(HL7524-7720) GND COOL ----- OFF

ENG 1 & 2 BLEED ----- OFF

APU BLEED ----- OFF

DITCHING pb ----- ON

Note : CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV NOT  
OPEN, COND LAV + GAL VENT FALUT -Disregard

ENG MASTER sw ----- OFF

SHUTDOWN CHECKLIST

"A/C PREPARED FOR SPRAYING"

UPON COMPLETE SPRAY (TIME CHK 1분)

용액과 마지막 용액 뿌린 시간 받고 적는다.

Holdover Time 결정!!!

(HL7524-7720) GND COOL ----- OFF

PITOT STATICS (GND Crew) ----- CHK

GND EQUIPMENT ----- REMOVE

DITCHING pb ----- OFF

OUTFLOW V/V - CHK OPEN (ECAM press)

TIME CHECK 1분후

ENG 1 & 2 BLEED ----- ON

BROADBAND XMT sw ----- ON

5분후

APU BLEED ----- ON

NORMAL PROCEDURE ----- RESUME

Note : INIT B check & re-enter

PREFLT CHKlist -> Req STARTUP -> CHKlist

BOTH ENGINES ARE STARTED

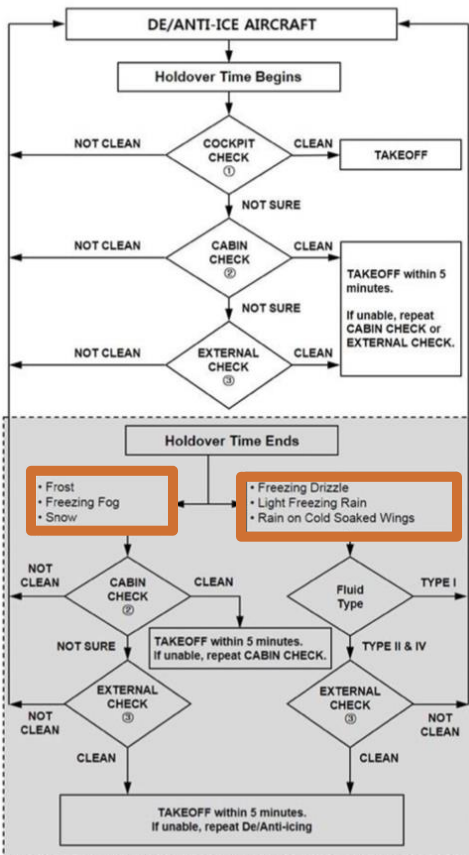
FLT CTL CHK, FLAPS UP, APU as REQ

DECISION TREE next page

# Cold Wx



# TAKEOFF DECISION TREE





# A330 PUS VOR 18L/R

ARRIVAL TO RKPK ↔  
 APPR VOR18R • VIA GAYHA STAR GAYHA3  
 APPR <VIAS TRANS KALOD

SEC ARRIVAL TO RKPK ↔  
 APPR 18R VIA NONE STAR V18R  
 TRANS NONE

Input PERF -> SEC COPY

CLR CP18L/R, DISCONT  
 MDA : [ ], L/D CONF chk

FIX : KMH 280(Base Turn), 284(Missed App)



## Missed App

Base Turn 이전 : L/H Turn **KMH 284** OUTBD  
 (PULL TRK)

Base Turn 이후 : Continue R/H Turn **KMH 284**  
 OUTBD (PULL TRK)



LOC 36 Circling  
 Next Page

# A330 PUS LOC 36L/R Circling 18L/R

## F-PLN

| APPR   | VIA   | STAR                      |
|--------|-------|---------------------------|
| LOC36L | KEVOX | KEVOX x<br>TRANS<br>MASTA |

## SEC F-PLN

| APPR  | STAR                                     |
|-------|------------------------------------------|
| 18L/R | V18L/R<br>(MDA: Blank)<br>(LDG CONF chk) |

**CI36L(CF36R) 3500 FI36L(FF36R) 2100**

**FIX : KMH 280(Base Turn), 310(Missed App)**



## Missed App

Base Turn 이전 : L/H Turn **KMH 310** OUTBD (PULL TRK)

Base Turn 이후 : Continue R/H Turn **KMH 310** OUTBD (PULL TRK)



# QRH LIST Reference Only

|       |                                                                                                                                                                                                                                          |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIR   | BLEED 1+2 FAULT(APU BLEED ON)<br>BLEED 1+2 FAULT(APU BLEED NOT AVAIL)                                                                                                                                                                    |
| DOOR  | COCKPIT DOOR FAULT                                                                                                                                                                                                                       |
| EIS   | DISPLAY UNIT FAILURE<br>EIS DISPLAY DISCREPANCY                                                                                                                                                                                          |
| ELEC  | C/B TRIPPED<br>ELEC EMER CONFIG SYS-REMAINING                                                                                                                                                                                            |
| ENG   | ALL ENG FAIL<br>ENGINE FUEL SYS CONTAMINATION<br>ENG RELIGHT IN FLIGHT<br>ENG STALL<br>ENGINE TAILPIPE FIRE<br>HIGH ENGINE VIBRATION<br>ON GND - NON ENG SHUTDN AFT ENG OFF<br>ONE ENGINE INOPER - CIRCLING APP<br>ENG1(2) EGT OVERLIMIT |
| F/CTL | LANDING WITH SLATS OR FLAPS JAMMED<br>NO FLAPS NO SLATS LANDING<br>RUDDER JAM / RUDDER PEDAL JAM<br>RUDDER TRIM RUNAWAY                                                                                                                  |
| FUEL  | FUEL IMBALANCE<br>FUEL LEAK<br>FUEL LOSS REDUCTION<br>FUEL OVERREAD<br>GRAVITY FUEL FEEDING<br>TRIM TANK FUEL UNUSABLE                                                                                                                   |
| L/G   | LANDING WITH ABNORMAL L/G<br>L/G GRAVITY EXTENSION                                                                                                                                                                                       |
| NAV   | ABNORMAL V ALPHA PROT<br>ADR 1+2+3 FAULT<br>UNREL SPD INDICATION<br>IR ALIGNMENT IN ATT MODE<br>NAV FM / GPS POS DISAGREE<br>ALL ADR OFF                                                                                                 |
| SMOKE | SMOKE / FUMES / AVNCS SMOKE<br>REMOVAL OF SMOKE / FUMES<br>SMOKE / FIRE FROM LITHIUM BATTERY                                                                                                                                             |
| WHEEL | WHEEL TIRE DAMAGE SUSPECTED                                                                                                                                                                                                              |
| MISC  | BOMB ON BOARD<br>COCKPIT WINDSHIELD/WINDOW ARCING<br>COCKPIT WINDSHIELD/WINDOW CRACKED<br>DITCHING<br>EMER EVAC<br>EMER LANDING<br>FORCED LANDING<br>OVERWEIGHT LANDING<br>SEVERE TURBULENCE<br>TAILSTRIKE VOLCANIC ASH ENCOUNTER        |



| SYSTEM RESET TABLE 1/2       |                      |                                                          |
|------------------------------|----------------------|----------------------------------------------------------|
| CKPT CTL pb 3초, RESET BTN 1초 |                      |                                                          |
| A-ICE                        | 2 WHC                | GND : A.ICE L(R)<br>WSHLD(WINDOW) HEAT                   |
| AIR                          | ENG BLEED pb         | AIR ENG 1(2) BLEED FAULT<br>GND :                        |
|                              | 2 PACK CONT          | AIR ENG 1(2) BLEED NOT CLSD<br>PACK CONT RESET           |
| COND                         | 1 ZONE CONT          | ZONE CONT RESET                                          |
| APU                          | APU MASTER           | Only APU SPD below 7%                                    |
| AUTO<br>FLT                  | REF FCOM             | GND/Before TAXI :<br>AUTO FLT A/THR OFF                  |
|                              | 2 FCU                | PART FCU, FCU                                            |
|                              | 2 FMGEC 2 FM         | AUTO FLT FM 1(2) FAULT<br>MAP NOT AVAIL                  |
|                              | MCDU OFF             | MCDU RESET                                               |
| BRKS                         | REF FCOM             | GND :<br>BRAKES RESIDUAL BRAKING<br>WHEEL N/W STRG FAULT |
| CAB PR                       | 2 CPC                | FAULTY CPC RESET                                         |
| COM                          | 2 CIDS               | Cabin Intercom Data Sys<br>RESET                         |
|                              | 2 ACP/AMU            | ACP RESET                                                |
|                              | RMP OFF              | RMP RESET                                                |
|                              | ELT RESET            | ELT RESET                                                |
| DATA<br>LINK                 | 1 ATSU               | INVALID DATA or ATSU RESET                               |
|                              | 1 AINS,1 CINS        | AINS, CINS RESET                                         |
| DOOR                         | 2 PSCU               | Proxi Switch Cont Unit RESET                             |
| EIS                          | DU OFF               | ECAM, PFD, ND RESET                                      |
|                              | 3 DMC                | DMC RESET                                                |
| ELEC                         | EXT A,B OFF          | GND : RESET EXT A,B                                      |
|                              | 3 BAT pb OFF         | BATT RESET                                               |
|                              | COMM, GALL<br>pb OFF | GND : COMMER, GALL RESET                                 |
| ENG                          | 2 EIVMU              | Eng Interface Vibration Monitor<br>Unit RESET            |
| F/CTL                        | PRIM,SEC OFF         | GND : PRIM, SEC RESET                                    |
|                              | 2 FCDC               | FCDC RESET                                               |
|                              | 2 FLAP               | GND : F/CTL FLAP SYS FAULT                               |
|                              | 2 SLAT               | GND : F/CTL SLAT SYS FAULT                               |



| SYSTEM RESET TABLE 2/2       |                        |                                      |
|------------------------------|------------------------|--------------------------------------|
| CKPT CTL pb 3초, RESET BTN 1초 |                        |                                      |
| FUEL                         | 2 FCMC                 | GND :<br>FUEL ZFW ZFCG DISAGREE      |
| FWS                          | 2 FWC, 2 SDAC          | Waring Sys RESET                     |
| L/G                          | 2 LGCIU                | LGCIU RESET                          |
| MISC                         | VSC<br>(Cabin aft C/B) | RESET VACUUM TOILET SYS<br>(Blue pb) |
| COND                         | 1 ZONE CONT            | ZONE CONT RESET                      |
| SMOKE                        | 2 SDCU                 | Smoke Dectect RESET                  |
| VENT                         | 2 VENT CONT            | RESET VENT CONT                      |
|                              | 1 AEVC                 | RESET VENT COMPUTER                  |



| GS KTS | KM   | MILES |
|--------|------|-------|
| 300    | 560  | 350   |
| 310    | 570  | 360   |
| 320    | 590  | 370   |
| 330    | 610  | 380   |
| 340    | 630  | 390   |
| 350    | 650  | 400   |
| 360    | 670  | 410   |
| 370    | 690  | 430   |
| 380    | 710  | 440   |
| 390    | 720  | 450   |
| 400    | 740  | 460   |
| 410    | 760  | 470   |
| 420    | 780  | 480   |
| 430    | 800  | 500   |
| 440    | 820  | 510   |
| 450    | 830  | 520   |
| 460    | 850  | 530   |
| 470    | 870  | 540   |
| 480    | 890  | 550   |
| 490    | 910  | 560   |
| 500    | 930  | 580   |
| 510    | 950  | 590   |
| 520    | 960  | 600   |
| 530    | 980  | 610   |
| 540    | 1000 | 620   |
| 550    | 1020 | 630   |
| 560    | 1040 | 650   |
| 570    | 1060 | 660   |
| 580    | 1070 | 670   |
| 590    | 1090 | 680   |
| 600    | 1110 | 690   |
| 610    | 1130 | 700   |
| 620    | 1150 | 710   |
| 630    | 1170 | 730   |
| 640    | 1190 | 740   |
| 650    | 1200 | 750   |
| 660    | 1220 | 760   |
| 670    | 1240 | 770   |
| 680    | 1260 | 780   |
| 690    | 1280 | 800   |
| 700    | 1300 | 810   |

